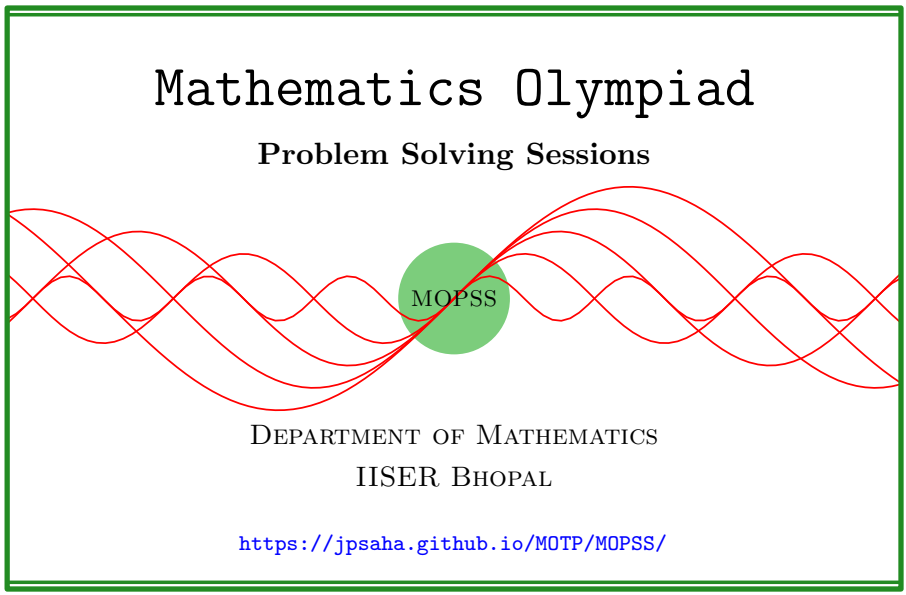
Congruences

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Congruences

§1.1 Infinitude of primes

Exercise 1 (Belarus National Olympiad 2024 Grade 11 Day 2 P6, , by I. Voronovich)**.** Let $2 = p_1 < p_2 < \cdots < p_n < \ldots$ be all prime numbers. Prove that for any positive integer $n \geq 3$ there exist at least $p_n + n - 1$ prime numbers, that do not exceed $p_1 p_2 \ldots p_n$.

Walkthrough —

(a) Consider the numbers

$$p_1, p_2, \ldots, p_{n-1},$$
$$p_1 p_2 \ldots p_{n-1} - 1, 2p_1 p_2 \ldots p_{n-1} - 1,$$
$$3p_1 p_2 \ldots p_{n-1} - 1, \ldots, p_n p_1 p_2 \ldots p_{n-1} - 1.$$

(b) Show that these are pairwise relatively prime.

Solution 1. Let $n \geq 3$ be an integer. Note that the integers

$$p_1 p_2 \ldots p_{n-1} - 1, 2p_1 p_2 \ldots p_{n-1} - 1, 3p_1 p_2 \ldots p_{n-1} - 1, \ldots, p_n p_1 p_2 \ldots p_{n-1} - 1$$

are pairwise relatively prime. Indeed, if a prime p divides two of them, say $ap_1 p_2 \ldots p_{n-1} - 1$ and $bp_1 p_2 \ldots p_{n-1} - 1$ with $1 \leq a < b \leq p_n$, then p divides their difference $(b-a)p_1 p_2 \ldots p_{n-1}$. Since p cannot be any of the primes $p_1, p_2, \ldots, p_{n-1}$, it follows that p divides $b - a$. Since $1 \leq b - a < p_n$, we conclude that $p < p_n$, that is, p is one of the primes $p_1, p_2, \ldots, p_{n-1}$. However, this is impossible since none of these primes divides $ap_1 p_2 \ldots p_{n-1} - 1$. Thus, the numbers

$$p_1 p_2 \ldots p_{n-1} - 1, 2p_1 p_2 \ldots p_{n-1} - 1, 3p_1 p_2 \ldots p_{n-1} - 1, \ldots, p_n p_1 p_2 \ldots p_{n-1} - 1$$

are pairwise relatively prime. Since $n \geq 3$, we obtain

$$p_1 p_2 \dots p_{n-1} - 1 \geq 2p_{n-1} - 1 > p_{n-1}.$$

This implies that the numbers

$$\begin{aligned} &p_1, p_2, \dots, p_{n-1}, \\ &p_1 p_2 \dots p_{n-1} - 1, 2p_1 p_2 \dots p_{n-1} - 1, \\ &3p_1 p_2 \dots p_{n-1} - 1, \dots, p_n p_1 p_2 \dots p_{n-1} - 1, \end{aligned}$$

are all distinct and greater than 1. Moreover, these $p_n + n - 1$ numbers are pairwise relatively prime. Therefore, each of them has at least one prime divisor, which does not divide any of the other numbers. Since all these numbers are less than or equal to $p_1 p_2 \dots p_n$, we conclude that there are at least $p_n + n - 1$ distinct prime numbers that do not exceed $p_1 p_2 \dots p_n$. ■

§1.2 Divisibility

Exercise 2 (Columbia National Olympiad 2025 P1, , proposed by Santiago Rodriguez)**.** Determine all permutations $a_1, a_2, \dots, a_{2025}$ of the numbers $1, 2, \dots, 2025$ such that the integer $a_j - a_i$ is divisible by $j - i$ for all $1 \leq i < j \leq 2025$.

Walkthrough —

(a) Let $a_1, a_2, \dots, a_{2025}$ be a permutation of the numbers $1, 2, \dots, 2025$ such that $a_j - a_i$ is divisible by $j - i$ for all $1 \leq i < j \leq 2025$. Show that if $a_1 < a_{2025}$, then $a_i = i$ holds for all $1 \leq i \leq 2025$.

(b) Show that the permutation $2026 - a_1, 2026 - a_2, \dots, 2026 - a_{2025}$ also satisfies the same divisibility condition.

(c) Conclude that the only two permutations that satisfy the divisibility condition are the identity permutation $1, 2, \dots, 2025$ and the permutation $2025, 2024, \dots, 1$.

(d) Note that if $a_1, a_2, \dots, a_{2025}$ is a permutation of the numbers $1, 2, \dots, 2025$ satisfying the divisibility condition, then so is the permutation $2026 - a_1, 2026 - a_2, \dots, 2026 - a_{2025}$.

(e) Conclude that the only two permutations that satisfy the divisibility condition are the identity permutation $1, 2, \dots, 2025$ and the permutation $2025, 2024, \dots, 1$.

Solution 2. Let $a_1, a_2, \dots, a_{2025}$ be a permutation of the numbers $1, 2, \dots, 2025$ such that $a_j - a_i$ is divisible by $j - i$ for all $1 \leq i < j \leq 2025$.

Claim — If $a_1 < a_{2025}$, then $a_i = i$ holds for any $1 \leq i \leq 2025$.

Proof of the Claim. Note that $a_{2025} - a_1$ is divisible by 2024. Using that $a_1 < a_{2025}$ and that $a_{2025} - a_1 \leq 2024$, we obtain that $a_1 = 1$ and $a_{2025} = 2025$. Let $1 \leq i \leq 2024$ be an integer such that $a_k = k$ for all $k \leq i$. If $i = 2024$, then we have that $a_{i+1} = a_{2025} = 2025 = i + 1$ and we are done. It remains to consider the case when $i < 2024$, which we assume from now on. Note that $2025 - a_{i+1} = a_{2025} - a_{i+1}$ is nonzero, and it is divisible by $2025 - i - 1$. Also note that $a_{i+1} \geq i + 1$. This yields

$$2025 - i - 1 \leq 2025 - a_{i+1} \leq 2025 - i - 1,$$

and thus $a_{i+1} = i + 1$. By induction, we conclude that $a_i = i$ for all $1 \leq i \leq 2025$. □

Note that

$$2026 - a_1, 2026 - a_2, \ldots, 2026 - a_{2025}$$

is also a permutation of the numbers $1, 2, \ldots, 2025$ and satisfies the same divisibility condition. By the above claim, we have that

$$a_i = i$$

holds for all $1 \leq i \leq 2025$, or that

$$2026 - a_i = i$$

holds for all $1 \leq i \leq 2025$. This proves that the permutation $a_1, a_2, \ldots, a_{2025}$ is either the identity permutation $1, 2, \ldots, 2025$ or the permutation $2025, 2024, \ldots, 1$. Since both of these permutations satisfy the divisibility condition, we conclude that these are the only two permutations that satisfy the condition. ■

§1.3 Factorization

Exercise 3 (Tournament of Towns 2021 Spring Junior A Level P1, , proposed by Mikhail Evdokimov)**.** The number $2021 = 43 \cdot 47$ is composite. Prove that if we insert any number of digits "8" between 20 and 21 then the number remains composite.

Exercise 4 (Putnam 2018 A1,)**.** Find all ordered pairs (a, b) of positive integers for which

$$\frac{1}{a} + \frac{1}{b} = \frac{3}{2018}.$$

Walkthrough —

(a) Rewrite the equation as

$$(3a - 2018)(3b - 2018) = 2018^2.$$

(b) Note that $2018 = 2 \times 1009$, where 1009 is a prime.

(c) Use congruence conditions modulo 3.

Solution 3.

■

Exercise 5 (Belarus National Olympiad 2025 Grade 11 Day 2 P5, , by V. Kamianetski)**.** Find the smallest positive integer n such that both $n^3 - n$ and $(n+1)^3 - (n+1)$ are divisible by 2025.

Walkthrough —

(a) Let n be a positive integer such that 2025 divides both $n^3 - n$ and $(n+1)^3 - (n+1)$. Show that $n(n+1)$ is divisible by 3^4 and 5^2.

(b) Conversely, show that if $n(n+1)$ is divisible by 3^4 and 5^2, then 2025 divides both $n^3 - n$ and $(n+1)^3 - (n+1)$.

(c) Find the smallest positive integer n such that $n(n+1)$ is divisible by 3^4 and 5^2.

Solution 4. Let n be a positive integer.

Assume that 2025 divides $n^3 - n$, and 2025 divides $(n+1)^3 - (n+1)$. It follows that 2025 divides the difference

$$((n+1)^3 - (n+1)) - (n^3 - n) = 3n^2 + 3n = 3n(n+1).$$

Note that $2025 = 3^4 \cdot 5^2$. Observe that 3^4 divides $3n(n+1)$, and hence, 3^3 divides $n(n+1)$. This shows that 3 does not divides $n - 1$. Since 3^4 divides

$$n^3 - n = n(n-1)(n+1),$$

it follows that 3^4 divides $n(n+1)$. Similarly, since 5^2 divides $3n(n+1)$, it follows that 5^2 divides $n(n+1)$.

Also note that if $n(n+1)$ is divisible by 3^4 and 5^2, then 2025 divides $n^3 - n$ and $(n+1)^3 - (n+1)$.

This shows that 2025 divides both $n^3 - n$ and $(n+1)^3 - (n+1)$ if and only if $n(n+1)$ is divisible by 3^4 and 5^2. Hence, we need to find the smallest positive integer n such that $n(n+1)$ is divisible by 3^4 and 5^2, or equivalently, one of the integers $n, n+1$ is divisible by 3^4, and one of the integers $n, n+1$ is divisible by 5^2.

The smallest positive integer n such that 3^4 divides n and 5^2 divides n is $3^4 \cdot 5^2$. The smallest positive integer n such that 3^4 divides n and 5^2 divides $n+1$ is $3^4 \cdot 4 = 324$. The smallest positive integer n such that 3^4 divides $n+1$ and 5^2 divides n is of the form $3^4 \cdot x - 1$ for some positive integer x such that 25 divides $3^4 \cdot x - 1$, and hence $x \equiv 21 \pmod{25}$, which shows that the smallest such positive integer n is $3^4 \cdot 21 - 1 = 680$. The smallest positive integer n such that 3^4 divides $n+1$ and 5^2 divides $n+1$ is $3^4 \cdot 5^2 - 1 = 2024$.

This shows that the smallest positive integer n such that 2025 divides both $n^3 - n$ and $(n+1)^3 - (n+1)$ is 324. ■

Exercise 6 (British Mathematical Olympiad Round 2 2015/16 P4)**.** Suppose that p is a prime number and that there are different positive integers u and v such that p^2 is the mean of u^2 and v^2. Prove that $2p - u - v$ is a square or twice a square.

> **Walkthrough** —
>
> **(a)**

Solution 5. Since p^2 is the mean of u^2 and v^2, we have

$$u^2 + v^2 = 2p^2.$$

Multiplying both sides by 2, we obtain

$$(u-v)^2 + (u+v)^2 = (2p)^2.$$

This yields

$$(2p - u - v)(2p + u + v) = (u - v)^2.$$

Let n be an odd positive integer dividing both $2p - u - v$ and $2p + u + v$. Then n divides their sum $4p$ and their difference $2(u+v)$. Since n is odd, it divides p and $u + v$. As p is prime, we have $n = 1$ or $n = p$.

Let us consider the case that $n = p$. Note that p divides $u + v$ and $u - v$. Since p is odd, it divides both u and v. Since u and v are distinct positive integers, this implies that

$$u^2 + v^2 > 2p^2,$$

which is a contradiction. This shows that $n = 1$, and hence, $2p - u - v$ and $2p + u + v$ admit no common odd positive integer divisor other than 1. It follows that there are nonnegative integers a and b, and relatively prime odd positive integers m and n, such that

$$2p - u - v = 2^a m, \quad 2p + u + v = 2^b n.$$

Since $2^{a+b}mn$ is a perfect square, we have that $a + b$ is even and both m and n are perfect squares. This proves that $2p - u - v$ is a square or twice a square. ■

Exercise 7 (British Mathematical Olympiad Round 1 2012/13 P4)**.** Find all positive integers n such that $12n-119$ and $75n-539$ are both perfect squares.

Walkthrough —

(a)

Solution 6. Let n be a positive integer such that $12n-119$ and $75n-539$ are both perfect squares. Let a, b be nonnegative integers such that

$$12n-119=a^2, \quad 75n-539=b^2.$$

This yields

$$\begin{aligned}(5a)^2-(2b)^2 &= 5^2\cdot 12n-5^2\cdot 119-2^2\cdot 75n+2^2\cdot 539\\ &= 2^2\cdot 420-119\cdot(25-4)\\ &= 21\cdot(80-119)\\ &= -21\cdot 39\\ &= -3^2\cdot 7\cdot 13,\end{aligned}$$

and this gives

$$(2b-5a)(2b+5a)=3^2\cdot 7\cdot 13.$$

Note that the factors on the left-hand side are both positive integers differing by a multiple of 10. Also note that $3, 13$ are congruent to $3 \pmod{10}$, and 7 is congruent to $-3 \pmod{10}$. This shows that $(2b-5a, 2b+5a)$ is equal to one of the following pairs:

$$(3, 3\cdot 7\cdot 13), (7, 3^2\cdot 13), (13, 3^2\cdot 7), (3\cdot 7\cdot 13, 3), (3^2\cdot 13, 7), (3^2\cdot 7, 13).$$

Since a is nonnegative, we have $2b+5a \geq 2b-5a$, so we only need to consider the first three cases. In the first case, note that 3 divides $10a$, and hence 3 divides a. In the second case, we have $a=11$. In the third case, we have $a=5$. Since $12n-119$ is equal to a^2, and 3 does not divide 119, we have that a is not divisible by 3, and hence a is equal to one of $11, 5$. If $a=11$, then

$$n=\frac{a^2+119}{12}=\frac{11^2+119}{12}=\frac{240}{12}=20,$$

and if $a=5$, then

$$n=\frac{a^2+119}{12}=\frac{5^2+119}{12}=\frac{144}{12}=12.$$

Observe that

$$12\cdot 20-119=121=11^2,$$

The content posted here and at this blog by Evan Chen are quite useful.

and

$$75 \cdot 20 - 539 = 1500 - 539 = 961 = 31^2$$

hold. Also note that

$$12 \cdot 12 - 119 = 25 = 5^2,$$

and

$$75 \cdot 12 - 539 = 900 - 539 = 361 = 19^2.$$

This proves that $n = 12, 20$ are the only positive integers such that both $12n - 119$ and $75n - 539$ are perfect squares. ■

Exercise 8 (British Mathematical Olympiad Round 1 2023 P4)**.** Find all positive integers n such that $n \times 2^n + 1$ is a perfect square.

Walkthrough —

(a) Write $n \times 2^n + 1 = k^2$ for some positive integer k.

(b) Rearranging gives $n \times 2^n = k^2 - 1 = (k-1)(k+1)$.

(c) Show that $k - 1$ and $k + 1$ are consecutive even integers.

(d) Using that one of $k - 1$ and $k + 1$ is not divisible by 4, show that one of $k - 1$ and $k + 1$ is divisible by 2^{n-1}.

(e) Deduce that $n \leq 4$.

Solution 7. Let n be a positive integer such that $n \times 2^n + 1 = k^2$ for some positive integer k. Rearranging gives

$$n \times 2^n = (k-1)(k+1).$$

Since the product $(k-1)(k+1)$ of the consecutive integers $k - 1$ and $k + 1$ is even, it follows that $k - 1, k + 1$ are both even. Thus, we can write $k - 1 = 2a$ and $k + 1 = 2b$ for some positive integers a and b with $b = a + 1$. Note that the greatest common divisor of the integers $k - 1$ and $k + 1$ is 2. Therefore, the greatest common divisor of the integers a and b is 1. This shows that at least one of the integers a and b is odd. Consequently, one of the integers $k - 1$ and $k + 1$ is divisible by 2^{n-1}. This shows that

$$n \times 2^n \geq 2^{n-1}(2^{n-1} - 2)$$

holds, which simplifies to $n \geq 2^{n-2} - 1$. Using induction, it follows that $2^{m-2} > m + 1$ holds for all integers $m \geq 5$. Therefore, we have $n \leq 4$. Note that n is not equal to any of the integers $1, 4$. Moreover, if $n = 2$, then $n \times 2^n + 1 = 9$, which is a perfect square. Finally, if $n = 3$, then $n \times 2^n + 1 = 25$, which is a perfect square. Thus, the only positive integer n such that $n \times 2^n + 1$ is a perfect square is $n = 2$ and $n = 3$. ■

Exercise 9 (British Mathematical Olympiad Round 2 2017/18 P3)**.** It is well known that, for each positive integer n,

$$1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2,$$

which is a perfect square. Determine whether or not there is a positive integer m such that

$$(m+1)^3 + (m+2)^3 + \cdots + (2m)^3$$

is a perfect square.

Walkthrough —

(a) Let us assume that there exists a positive integer m such that the sum $(m+1)^3 + (m+2)^3 + \cdots + (2m)^3$ is a perfect square.

(b) Considering congruences modulo 4, show that m must be odd.

(c) Write $m = 2n+1$ for some non-negative integer n.

(d) Consider the greatest common divisor of $3n+2$ and $5n+4$.

(e) Using congruences modulo 3 and 5, obtain a contradiction to show that there is no such positive integer m.

Solution 8. Let m be a positive integer such that

$$(m+1)^3 + (m+2)^3 + \cdots + (2m)^3$$

is a perfect square. Write

$$(m+1)^3 + (m+2)^3 + \cdots + (2m)^3 = k^2$$

for some positive integer k. Using the formula for the sum of cubes, we have

$$\begin{aligned}
k^2 &= (m+1)^3 + (m+2)^3 + \cdots + (2m)^3 \\
&= (1^3 + 2^3 + \cdots + (2m)^3) - (1^3 + 2^3 + \cdots + m^3) \\
&= \left(\frac{2m(2m+1)}{2}\right)^2 - \left(\frac{m(m+1)}{2}\right)^2 \\
&= (m(2m+1))^2 - \left(\frac{m(m+1)}{2}\right)^2 \\
&= m^2\left(2m+1-\frac{m+1}{2}\right)\left(2m+1+\frac{m+1}{2}\right) \\
&= \frac{m^2}{4}(3m+1)(5m+3).
\end{aligned}$$

This shows that

$$(3m+1)(5m+3)$$

is a perfect square. Since $(3m+1)(5m+3)$ is congruent to $-(m-1)^2$ modulo 4, it follows that m is odd. Write $m = 2n+1$ for some non-negative integer n. We obtain

$$k^2 = m^2(3n+2)(5n+4).$$

Note that the greatest common divisor of $3n+2$ and $5n+4$ divides $3(5n+4) - 5(3n+2) = 2$. This shows that the integers $3n+2$ and $5n+4$ are the squares of two coprime positive integers, or, these two integers are twice the squares of two coprime positive integers. In the first case, there exist coprime positive integers a and b such that

$$3n+2 = a^2, \quad 5n+4 = b^2.$$

Eliminating n gives

$$5a^2 - 3b^2 = -2.$$

Reducing this equation modulo 3 gives $a^2 \equiv 2 \pmod 3$, which is impossible. In the second case, there exist coprime positive integers a and b such that

$$3n+2 = 2a^2, \quad 5n+4 = 2b^2.$$

Eliminating n gives

$$5a^2 - 3b^2 = -1.$$

Reducing this equation modulo 5 gives $3b^2 \equiv 1 \pmod 5$, which implies $b^2 \equiv 2 \pmod 5$, which is impossible. This shows that there is no positive integer m such that

$$(m+1)^3 + (m+2)^3 + \cdots + (2m)^3$$

is a perfect square. ■

Exercise 10 (British Mathematical Olympiad Round 1 2006/7 P6, Moscow Mathematical Olympiad 2024 Grade 11 Day 2 P3 by T. A. Garmanova, folklore?)**.** Let n be an integer. Show that, if $2 + 2\sqrt{1+12n^2}$ is an integer, then it is a perfect square.

> **Walkthrough** —
>
> **(a)**

Solution 9. Assume that $2 + 2\sqrt{1+12n^2}$ is equal to an integer k. Then we have

$$4(1+12n^2) = (k-2)^2.$$

This shows that $k-2$ is even, and hence so is k. Let $k = 2m$ for some integer m. We obtain

$$1 + 12n^2 = (m-1)^2,$$

which can be rearranged to give

$$12n^2 = m^2 - 2m.$$

This shows that m is even. Write $m = 2r$ for some integer r. Then we have

$$3n^2 = r^2 - r = r(r-1).$$

Note that r and $r-1$ are consecutive integers, and hence they are coprime. If 3 divides r, then from the equation above, and that $r, r-1$ are coprime, it follows that $r-1$ is a perfect square, which implies -1 is a quadratic residue modulo 3, a contradiction. Therefore, 3 does not divide r.

Using

$$3n^2 = r(r-1)$$

and that 3 does not divide r, it follows that r is a perfect square[1]. Since

$$k = 2m = 4r,$$

we conclude that k is a perfect square. This implies that $2 + 2\sqrt{1 + 12n^2}$ is a perfect square. ■

Exercise 11 (British Mathematical Olympiad Round 2 2008/9 P1)**.** Find all solutions in non-negative integers a, b to

$$\sqrt{a} + \sqrt{b} = \sqrt{2009}.$$

Walkthrough —

(a)

Solution 10. Let a, b be nonnegative integers satisfying $\sqrt{a} + \sqrt{b} = \sqrt{2009}$. Note that

$$\sqrt{a} = \sqrt{2009} - \sqrt{b}$$

holds. Squaring both sides, we get

$$a = 2009 + b - 2\sqrt{2009b}.$$

This gives

$$4 \cdot 2009b = (2009 + b - a)^2.$$

This implies that 4 divides $(2009 + b - a)^2$, and hence 2 divides $2009 + b - a$. It follows that $2009b$ is a perfect square. Since $2009 = 7^2 \cdot 41$, we obtain that $41b$ is a perfect square. Similarly, it also follows that $41a$ is a perfect square. Thus, there exist nonnegative integers m, n such that

$$a = 41m^2, \quad b = 41n^2.$$

[1] How does it follow?

Substituting these into the original equation, we get

$$\sqrt{41}m + \sqrt{41}n = \sqrt{2009} = 7\sqrt{41},$$

which simplifies to

$$m + n = 7.$$

Moreover, if m, n are nonnegative integers satisfying $m + n = 7$, then

$$\sqrt{41m^2} + \sqrt{41n^2} = \sqrt{41}(m + n) = 7\sqrt{41} = \sqrt{2009}$$

holds. Therefore, the solutions in nonnegative integers (a, b) to the original equation are given by

$$(a, b) = (41m^2, 41n^2)$$

for nonnegative integers m, n satisfying $m + n = 7$. Equivalently, the solutions are given by

$$(a, b) = (41m^2, 41(7 - m)^2)$$

for $m = 0, 1, 2, \ldots, 7$. ■

Exercise 12 (IOQM 2023 P11,). A positive integer m has the property that m^2 is expressible in the form $4n^2 - 5n + 16$ where n is an integer (of any sign). Find the maximum possible value of $|m - n|$.

Summary — Write the equation $m^2 = 4n^2 - 5n + 16$ as a quadratic polynomial in n. Note that its discriminant is a perfect square. Obtain a suitable factorization.

Walkthrough —

(a) Let m, n be integers satisfying $m^2 = 4n^2 - 5n + 16$, or equivalently,

$$4n^2 - 5n + (16 - m^2) = 0.$$

(b) Since n is an integer, the discriminant of the quadratic polynomial $4n^2 - 5n + (16 - m^2)$ in n is a perfect square, that is, there exists a nonnegative integer ℓ such that

$$25 - 16(16 - m^2) = \ell^2,$$

or equivalently, $16m^2 - \ell^2 = 231$, which gives

$$(4m - \ell)(4m + \ell) = 3 \cdot 7 \cdot 11.$$

(c) Note that $4m - \ell, 4m + \ell$ are odd and hence congruent to $1, -1$ modulo 4 (in some order). Also note that $4m - \ell < 4m + \ell$ holds.

(d) This shows that $(4m - \ell, 4m + \ell)$ is equal to one of

$$(1, 3 \cdot 7 \cdot 11), (3, 77), (7, 33), (11, 21),$$

and consequently, $8m$ is equal to one of $232, 80, 40, 32$, which shows that m is equal to one of $29, 10, 5, 4$.

(e) If $m = 29$, then $4n^2 - 5n - 825 = 0$, which gives $n = 15$.

(f) If $m = 10$, then $4n^2 - 5n - 84 = 0$ holds, which gives $n = -4$.

(g) If $m = 5$, then $4n^2 - 5n - 9 = 0$ holds, implying $n = -1$.

(h) If $m = 4$, then $n = 0$.

(i) Conclude that the maximum possible value of $|m - n|$ is equal to 14.

Solution 11. Let m be a positive integer and n be an integer satisfying

$$m^2 = 4n^2 - 5n + 16,$$

or equivalently, the equation

$$4n^2 - 5n + (16 - m^2) = 0.$$

Since n is an integer, the discriminant of the quadratic polynomial $4x^2 - 5x + (16 - m^2)$ is a perfect square, that is, there exists a nonnegative integer ℓ such that

$$25 - 16(16 - m^2) = \ell^2,$$

or equivalently,

$$16m^2 - \ell^2 = 231$$

holds. This yields

$$(4m - \ell)(4m + \ell) = 3 \cdot 7 \cdot 11.$$

Note that ℓ is an odd integer, and hence $4m - \ell, 4m + \ell$ are odd and congruent to $1, -1$ modulo 4 (in some order). Using $4m - \ell < 4m + \ell$, we obtain $(4m - \ell, 4m + \ell)$ is equal to one of

$$(1, 3 \cdot 7 \cdot 11), (3, 77), (7, 33), (11, 21).$$

This shows that $8m$ is equal to one of $232, 80, 40, 32$, which gives that m is equal to one of $29, 10, 5, 4$.

If $m = 29$, then

$$4n^2 - 5n - 825 = 0$$

holds, which gives $n = 15$. If $m = 10$, then

$$4n^2 - 5n - 84 = 0$$

holds, which implies $n = -4$. If $m = 5$, then

$$4n^2 - 5n - 9 = 0$$

holds, which yields $n = -1$. If $m = 4$, then

$$4n^2 - 5n = 0$$

holds, which gives $n = 0$.

Note that the pairs

$$(29, 15), (10, -4), (5, -1), (4, 0)$$

also satisfy the given equation. Hence, the maximum possible value of $|m - n|$ is

$$\max\{29 - 15, 10 - (-4), 5 - (-1), 4 - 0\} = \max\{14, 14, 6, 4\} = 14.$$

■

Exercise 13 (SMMC 2017 B2,). Determine all pairs (p, q) of positive integers such that p and q are primes, and $p^{q-1} + q^{p-1}$ is the square of an integer.

Walkthrough —

(a) Check that $(p, q) = (2, 2)$ works.

(b) If p, q are odd, show that $p^{q-1} + q^{p-1} \equiv 2 \pmod 4$, so it cannot be a perfect square.

(c) Without loss of generality, let p be an odd prime, and $q = 2$. Then $p^{q-1} + q^{p-1} = 2^{p-1} + p$.

(d) Write $2^{p-1} + p = m^2$.

(e) Factorize to obtain $(m - 2^{\frac{p-1}{2}})(m + 2^{\frac{p-1}{2}}) = p$, which gives

$$m + 2^{\frac{p-1}{2}} = p.$$

(f) Note that $p < 2^{\frac{p-1}{2}}$ holds[a] for $p \geq 7$.

(g) Conclude that p can only be 3 or 5.

[a] Try to show that $2^{\frac{n-1}{2}} > n$ holds for all integers $n \geq 7$.

Solution 12. First, note that $(p, q) = (2, 2)$ works since $2^{2-1} + 2^{2-1} = 4 = 2^2$.

If p and q are odd primes, then both $p - 1$ and $q - 1$ are even, and hence,

$$p^{q-1} + q^{p-1} \equiv 1 + 1 \equiv 2 \pmod 4$$

holds. Since no perfect square is congruent to 2 modulo 4, it follows that not both of p and q are odd.

Without loss of generality[2], let p be an odd prime, and $q = 2$. Then

$$p^{q-1} + q^{p-1} = p^{2-1} + 2^{p-1} = p + 2^{p-1}.$$

[2] Why there is no loss of generality?

Write

$$p + 2^{p-1} = m^2$$

for some positive integer m. Rearranging gives

$$m^2 - 2^{p-1} = p,$$

which yields the factorization

$$(m - 2^{\frac{p-1}{2}})(m + 2^{\frac{p-1}{2}}) = p.$$

Since p is prime and

$$m - 2^{\frac{p-1}{2}} < m + 2^{\frac{p-1}{2}},$$

we obtain that

$$m + 2^{\frac{p-1}{2}} = p,$$

Using the following claim, we conclude that p can only be 3 or 5. If $p = 3$, then $p^{q-1} + q^{p-1} = 3^{2-1} + 2^{3-1} = 3 + 4 = 7$. If $p = 5$, then $p^{q-1} + q^{p-1} = 5^{2-1} + 2^{5-1} = 5 + 16 = 21$. Thus, the only solution is $(p, q) = (2, 2)$.

Claim — For any positive integer $n \geq 7$, the inequality

$$2^{\frac{n-1}{2}} > n$$

holds.

Proof. We prove this by induction. For $n = 7$, we have $2^{\frac{7-1}{2}} = 2^3 = 8 > 7$. Now, assume that the inequality holds for some $k \geq 7$. Note that

$$2^{\frac{(k+1)-1}{2}} = 2^{\frac{k-1}{2}+\frac{1}{2}} = \sqrt{2} \cdot 2^{\frac{k-1}{2}} > \sqrt{2} \cdot k$$

holds. Observe that

$$\begin{aligned}(\sqrt{2}k)^2 - (k+1)^2 &= 2k^2 - (k^2 + 2k + 1) \\ &= k^2 - 2k - 1 \\ &= (k-1)(k-3) - 2 \\ &\geq (7-1)(7-3) - 2 \\ &> 0.\end{aligned}$$

This shows that

$$2^{\frac{(k+1)-1}{2}} > \sqrt{2} \cdot k > k + 1.$$

Thus, by induction, the claim holds for all integers $n \geq 7$. □

■

§1.4 Completing squares, factorizing

Example 14. Find all solutions of $x^2 + 3y^2 = 4$ in integers. Use it to find all solutions of $m^2 + mn + n^2 = 1$ in integers.

> **Walkthrough** — Observe that
>
> $$4(m^2 + mn + n^2) = (2m + n)^2 + 3n^2.$$

Exercise 15 (BStatBMath 2013 P7)**.** Let N be a positive integer such that $N(N - 101)$ is the square of a positive integer. Find all possible values of N. (Note that 101 is a prime number).

Solution 13. Let N be a positive integer such that for some positive integer k,

$$N(N - 101) = k^2$$

holds. This gives

$$(2N - 101)^2 = 4k^2 + 101^2,$$

which yields

$$(2N - 101 - 2k)(2N - 101 + 2k) = 101^2.$$

Using $N(N - 101) = k^2$, $N \geq 1$ and $k \geq 1$, it follows that $N > 101$. This shows that $2N - 101 - 2k, 2N - 101 + 2k$ are distinct positive divisors of 101^2. This implies that $2N - 101 - 2k, 2N - 101 + 2k$ are equal to $1, 101^2$ in some order, which yields

$$N = \frac{1 + 101^2 + 2 \cdot 101}{4} = \frac{102^2}{4} = 51^2 = 2601.$$

Note that

$$2601(2601 - 101) = 51^2 \times 50^2,$$

which is a perfect square. This proves that $N = 2601$ is a only solution. ■

Exercise 16 (British Mathematical Olympiad Round 1 2011/12 P1)**.** Find all (positive or negative) integers n for which $n^2 + 20n + 11$ is a perfect square. *Remember that you must justify that you have found them all.*

> **Walkthrough** —
>
> **(a)**

Solution 14. Let n be an integer such that $n^2 + 20n + 11$ is a perfect square. Then there exists a nonnegative integer m such that

$$n^2 + 20n + 11 = m^2.$$

Note that

$$(n+10)^2 - 89 = m^2$$

holds, which yields

$$(n+10-m)(n+10+m) = 89.$$

Since 89 is prime and the integer m is nonnegative, it follows that

$$(n+10-m, n+10+m)$$

is equal to one of $(1, 89)$ or $(-89, -1)$. In the first case, we have

$$n = 35, m = 44,$$

and in the second case, we have

$$n = -55, m = 44.$$

Note that both $n = 35$ and $n = -55$ satisfy the condition that $n^2 + 20n + 11$ is a perfect square. Therefore, the integers n for which $n^2 + 20n + 11$ is a perfect square are

$$35, -55.$$

■

Exercise 17 (USAJMO 2015 P2)**.** Solve in integers the equation

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3.$$

Walkthrough —

(a)

Solution 15. Let x, y be integers satisfying the equation. Rearranging, we have

$$(x+y)^2 - xy = \left(\frac{x+y}{3} + 1\right)^3.$$

Note that 3 divides $x + y$. Write $x + y = 3n$ for some integer n. We obtain

$$9n^2 - xy = (n+1)^3,$$

which yields

$$9n^2 - x(3n - x) = n^3 + 3n^2 + 3n + 1,$$

which simplifies to

$$x^2 - 3nx = n^3 - 6n^2 + 3n + 1.$$

This gives

$$4x^2 - 12nx + 9n^2 = 4n^3 - 15n^2 + 12n + 4,$$

which shows that

$$(2x - 3n)^2 = (n - 2)(4n^2 - 7n - 2),$$

and hence,

$$(2x - 3n)^2 = (n - 2)^2(4n + 1)$$

holds. This implies that $4n + 1$ is a perfect square. Write $4n + 1 = (2m + 1)^2$ for some integer m. This gives $n = m^2 + m$, and hence we obtain

$$(2x - 3n)^2 = (n - 2)^2(2m + 1)^2,$$

which implies that $2x - 3n$ equals $\pm(n - 2)(2m + 1)$. This shows that

$$\begin{aligned} x &= \frac{3n + n(2m + 1)}{2} - 2m - 1 \\ &= 2n + mn - 2m - 1 \\ &= 2m^2 + 2m + m^3 + m^2 - 2m - 1 \\ &= m^3 + 3m^2 - 1, \end{aligned}$$

holds or

$$\begin{aligned} x &= \frac{3n - n(2m + 1)}{2} + 2m + 1 \\ &= n - mn + 2m + 1 \\ &= m^2 + m - m^3 - m^2 + 2m + 1 \\ &= -m^3 + 3m + 1 \end{aligned}$$

holds. It follows that (x, y) is equal to one of

$$(m^3 + 3m^2 - 1, -m^3 + 3m + 1), \quad (-m^3 + 3m + 1, m^3 + 3m^2 - 1).$$

Also note that if m, n are integers satisfying $n = m^2 + m$, and (x, y) is a pair of integers satisfying $x + y = 3n$ and

$$(2x - 3n)^2 = (n - 2)^2(2m + 1)^2,$$

then it follows that

$$x^2 + xy + y^2 = \left(\frac{x + y}{3} + 1\right)^3.$$

Consequently, the integer solutions to the equation are exactly the pairs of the form

$$(m^3 + 3m^2 - 1, -m^3 + 3m + 1), \quad (-m^3 + 3m + 1, m^3 + 3m^2 - 1),$$

for some integer m. ■

Exercise 18 (British Mathematical Olympiad Round 1 2016/17 P3)**.** Determine all pairs (m, n) of positive integers which satisfy the equation $n^2 - 6n = m^2 + m - 10$.

Walkthrough —

(a) Complete the square on both sides, and rearrange to get a suitable factorization.

Solution 16. Let m, n be positive integers satisfying

$$n^2 - 6n = m^2 + m - 10.$$

This yields

$$(n-3)^2 = m^2 + m - 1,$$

which implies

$$(n-3)^2 = \left(m + \frac{1}{2}\right)^2 - \frac{5}{4}.$$

Rearranging gives

$$(2m+1)^2 - (2(n-3))^2 = 5.$$

Factorizing yields

$$(2m + 1 - 2(n-3))(2m + 1 + 2(n-3)) = 5.$$

Using $n^2 - 6n = m^2 + m - 10$, it follows that n is not an odd integer. Note that if $n = 2$, then the given equation holds yields $m = 1$. It remains to consider the case that $n \geq 4$, which we assume from now on. Note that

$$2m + 1 - 2(n-3) \leq 2m + 1 + 2(n-3)$$

holds, and hence we obtain

$$2m + 1 - 2(n-3) = 1, \quad 2m + 1 + 2(n-3) = 5.$$

This gives $m = 1, n = 4$.

Moreover, if $m = 1, n = 4$, then $n^2 - 6n = m^2 + m - 10$ holds. Further, if $m = 1, n = 2$, then also the given equation holds.

Therefore, the pairs (m, n) of positive integers which satisfy the equation $n^2 - 6n = m^2 + m - 10$ are precisely $(1, 2)$ and $(1, 4)$. ■

Exercise 19 (RMO 2024a P1,). Find all positive integers x, y such that $202x + 4x^2 = y^2$.

Solution 17. Let x, y be positive integers satisfying $202x + 4x^2 = y^2$. Note that y^2 is even, and hence, so is y. This shows that 4 divides $y^2 - 4x^2 = 202x$. It follows that x is even.

Note that

$$4y^2 = (4x)^2 + 2 \cdot 101 \cdot (4x)$$

holds, which gives

$$101^2 = (4x + 101 - 2y)(4x + 101 + 2y). \tag{1}$$

Since x, y are positive integers, it follows that $4x + 101 - 2y < 4x + 101 + 2y$. This shows that

$$4x + 101 - 2y = 1, \quad 4x + 101 + 2y = 101^2, \tag{2}$$

which yields

$$x = 1250, \quad y = 2550.$$

Also note that if $x = 1250$ and $y = 2550$, then Eq. (2) holds, which yields Eq. (1), which implies $202x + 4x^2 = y^2$. Hence, the solution to the given equation in the positive integers is precisely

$$x = 1250, \quad y = 2550.$$

■

§1.5 Warm up

Exercise 20. Show that the square of no rational number is equal to 2.

> **Walkthrough** —
>
> **(a)** On the contrary, suppose there exist positive integers a, b such that $(a/b)^2 = 2$ and $\gcd(a, b) = 1$.
>
> **(b)** This gives $a^2 - 2b^2 = 0$.
>
> **(c)** Read it modulo 3, and conclude.

Solution 18. On the contrary, let us assume that there exist positive integers a, b such that

$$(a/b)^2 = 2 \quad \text{and} \quad \gcd(a, b) = 1.$$

This gives $a^2 - 2b^2 = 0$. Note that the square of any integer is congruent to either 0 or 1 modulo 3. Since 3 does not divide both of a and b, it follows that $a^2 - 2b^2$ is congruent to either 1 or 2 modulo 3, which is a contradiction. ■

Exercise 21 (Japan Mathematical Olympiad 1992 P1)**.** Let x, y be coprime positive integers with $xy > 1$, and let n be an even positive integer. Prove that $x^n + y^n$ is not divisible by $x + y$.

Walkthrough —

(a) Assume that $x+y$ divides x^n+y^n.

(b) Since x is congruent to $-y$ modulo $x+y$, and n is even, it follows that $x^n+y^n \equiv 2y^n \pmod{x+y}$.

(c) Conclude that $x+y$ divides 2.

Solution 19. On the contrary, let us assume that $x+y$ divides x^n+y^n. Since x is congruent to $-y$ modulo $x+y$, and n is even, it follows that x^n is congruent to y^n modulo $x+y$. This shows that x^n+y^n is congruent to $2y^n$ modulo $x+y$. Therefore, $x+y$ divides $2y^n$. Since x, y are coprime, it follows that $x+y$ and y^n are coprime. Hence, $x+y$ divides 2. Since x, y are positive integers with $xy > 1$, it follows that $x+y \geq 3$, which is a contradiction. Thus, $x+y$ does not divide x^n+y^n. ■

Exercise 22 (Hungarian Mathematical Olympiad 2005/06 Specialized Math Schools P2)**.** Denote by $d(n)$ the number of positive divisors of n. Suppose that r and s are positive integers with the property that $d(ks) \geq d(kr)$ for each $k \in \mathbb{N}$. Prove that r divides s.

Walkthrough —

(a) Note that it suffices to consider the case when r is a power of a prime, say $r = p^a$ for some prime p, and some positive integer a, which we assume from now on.

(b) Observe that $s > 1$ since $d(s) \geq d(p^a) \geq 2$. Write

$$s = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_t^{\alpha_t},$$

where t is a positive integer, $p_1, p_2, \dots, p_t$ are distinct primes, and $\alpha_1, \alpha_2, \dots, \alpha_t$ are positive integers.

(c) If $p \neq p_i$ for all $i \in \{1, 2, \dots, t\}$, then, take $k = p^u$ for some suitable positive integer u and obtain a contradiction.

(d) If $p = p_i$ for some $i \in \{1, 2, \dots, t\}$, then, take $k = p^v$ for some suitable positive integer v and show that $a \leq \alpha_i$.

Solution 20. Note that it suffices to consider the case when r is a power of a prime, say $r = p^a$ for some prime p, and some positive integer a. Indeed, if x is a prime power dividing r, then for each positive integer k, we have

$$d(ks) \geq d(kr) \geq d(kx),$$

and if we can show that every prime power dividing r also divides s, then it follows that r divides s.

Observe that $s > 1$ since $d(s) \geq d(p^a) \geq 2$. Write

$$s = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_t^{\alpha_t},$$

where t is a positive integer, $p_1, p_2, \dots, p_t$ are distinct primes, and $\alpha_1, \alpha_2, \dots, \alpha_t$ are positive integers.

Let us first show that $p = p_i$ for some $i \in \{1, 2, \dots, t\}$. Suppose, for the sake of contradiction, that $p \neq p_i$ for all $i \in \{1, 2, \dots, t\}$. Then, taking $k = p^u$ with

$$u = a(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_t + 1) - a,$$

we have

$$\begin{aligned} d(ks) &= (u+1)(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_t + 1), \\ d(kr) &= d(p^{u+a}) \\ &= u + a + 1. \end{aligned}$$

Since $u+a+1$ is coprime to $(\alpha_1+1)(\alpha_2+1)\dots(\alpha_t+1)$, it follows that $u+a+1$ divides $u+1$. This is a contradiction since $a \geq 1$. Thus, $p = p_i$ for some $i \in \{1, 2, \dots, t\}$. Renaming indices if necessary, we may assume that $i = 1$. Taking $k = p^u$ with

$$u = a(\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_t + 1) - a,$$

we have

$$\begin{aligned} d(ks) &= (u + \alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_t + 1), \\ d(kr) &= d(p^{u+a}) \\ &= u + a + 1. \end{aligned}$$

Since $u + a + 1$ is coprime to $(\alpha_2 + 1) \dots (\alpha_t + 1)$, it follows that $u + a + 1$ divides $u + \alpha_1 + 1$. This shows that $a \leq \alpha_1$. ■

Exercise 23 (All-Russian Mathematical Olympiad 2018 Grade 9 Day 1 P1, , by A. Golovanov)**.** Let $a_1, a_2, \dots$ be an infinite strictly increasing sequence of positive integers, and $p_1, p_2, \dots$ be an infinite strictly increasing sequence of primes such that p_n divides a_n for all $n \geq 1$. Assume that $a_n - a_k = p_n - p_k$ for all $n, k \geq 1$. Prove that the sequence $a_1, a_2, \dots$ consists only of prime numbers.

Walkthrough —

(a) Show that p_n divides $p_1 - a_1$ for all $n \geq 1$.

(b) Prove that a_n is equal to p_n for all $n \geq 1$.

Solution 21. Note that

$$a_n - p_n = a_1 - p_1$$

holds for all $n \geq 1$. It follows that p_n divides $a_1 - p_1$ for all $n \geq 1$. This implies that $a_1 = p_1$. Thus, we have $a_n = p_n$ for all $n \geq 1$, which completes the proof. ■

Exercise 24 (All-Russian Mathematical Olympiad 2024 Grade 10 Day 1 P1, , by A. Kuznetsov)**.** Let p and q be different prime numbers. We are given an infinite decreasing arithmetic progression in which each of the numbers p^{23}, p^{24}, q^{23} and q^{24} occurs. Show that the numbers p and q also occur in this progression.

> **Walkthrough** — Let d denote the common difference of the arithmetic progression.
>
> **(a)** Using that d divides $p^{24} - p^{23}$, show that p does not divide d.
>
> **(b)** Using that d divides $p^{24} - p^{23}$, show that d divides $p - 1$.
>
> **(c)** Show that p occurs in the arithmetic progression.

Solution 22. Denote the common difference of the arithmetic progression by d. Since p, q are distinct primes, it follows that p does not divide $p^{23} - q^{23}$. Using that d divides $p^{23} - q^{23}$, we conclude that p does not divide d. Note that d also divides $p^{24} - p^{23}$, which implies that d divides $p - 1$. It follows that d divides $p^{23} - p$. Since d is negative, we conclude that p occurs in the arithmetic progression. By a similar argument, it follows that q also occurs in the arithmetic progression. ■

Exercise 25 (Mexico National Olympiad 2013 P1,)**.** All prime numbers are written in order, $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$. Find all pairs of positive integers a and b with $a - b \geq 2$, such that $p_a - p_b$ divides $2(a - b)$.

> **Walkthrough** —
>
> **(a)**

Solution 23. Let a, b be positive integers such that $a - b \geq 2$ and $p_a - p_b$ divides $2(a - b)$.

Let us consider the case that $b \geq 2$. Note that p_i, p_{i+1} differ by at least 2 for all $i \geq 2$, since all primes greater than 2 are odd. This gives

$$p_a - p_b = (p_a - p_{a-1}) + (p_{a-1} - p_{a-2}) + \cdots + (p_{b+1} - p_b) \geq 2(a - b).$$

Since $p_a - p_b$ divides $2(a - b)$, we must have $p_a - p_b = 2(a - b)$, implying that p_i, p_{i+1} differ by exactly 2 for all $i = b, b + 1, \ldots, a - 1$. Since $a \geq b + 2$, it

follows that the primes p_b, p_{b+1}, p_{b+2} are three consecutive primes differing by 2. Note that p_b, p_{b+1}, p_{b+2} are congruent to $0, 1, 2$ modulo 3 in some order. Thus, one of these three primes is divisible by 3. We obtain $p_b = 3$, which gives $b = 2$. If $a \geq b + 3$, then $p_{b+3} - p_{b+2} \geq 4$ holds, which implies

$$p_a - p_b = (p_a - p_{a-1}) + (p_{a-1} - p_{a-2}) + \cdots + (p_{b+1} - p_b) \geq 2(a - b) + 2.$$

This contradicts the fact that $p_a - p_b \leq 2(a - b)$. It follows that $a \leq b + 2$. Also note that $a \geq b + 2$. Thus, we conclude that $a = b + 2 = 4$.

It remains to consider the case $b = 1$, which we assume from now on. We have $p_a - p_1 = p_a - 2$ divides $2(a - 1)$. Since p_a is odd, it follows that $p_a - 2$ is odd, and hence it divides $a - 1$. This implies that $p_a - 2 \leq a - 1$, or equivalently $p_a \leq a + 1$. Using $a \geq b + 2 = 3$, we obtain

$$p_a = (p_a - p_{a-1}) + \cdots + (p_3 - p_2) + p_2 \geq 2(a - 2) + 3 = 2a - 1.$$

Combining the two inequalities, we get $2a - 1 \leq a + 1$, or equivalently $a \leq 2$, contradicting $a \geq 3$.

This proves that (a, b) is equal to $(4, 2)$. Note that this pair indeed satisfies the condition since $p_4 - p_2 = 7 - 3 = 4$ divides $2(4 - 2) = 4$. Consequently, the only solution is $(a, b) = (4, 2)$. ■

Exercise 26 (IOQM 2023 P4,). Let x, y be positive integers such that

$$x^4 = (x - 1)(y^3 - 23) - 1.$$

Find the maximum possible value of $x + y$.

Summary — Look for some divisibility conditions.

Walkthrough —

(a) Note that $x \neq 1$ and $x - 1$ divides 2. It follows that $x = 2$ or $x = 3$. If $x = 2$, then $y^3 = 23 + 2^4 + 1 = 40$, which is impossible since y is an integer.

(b) If $x = 3$, then $y^3 = 23 + \frac{1}{2}(3^4 + 1) = 23 + 41 = 64$, which gives $y = 4$.

(c) It follows that $x + y = 7$.

Solution 24. Let x, y be positive integers satisfying the given equation. Note that $x \geq 2$. Using the congruence $x \equiv 1 \pmod{x - 1}$, it follows that $x - 1$ divides 2. This shows that $x = 2$ or $x = 3$. If $x = 2$, then we obtain

$$y^3 = 23 + 2^4 + 1 = 40,$$

which is impossible since y is an integer. If $x = 3$, then we obtain

$$y^3 = 23 + \frac{1}{2}(3^4 + 1) = 23 + 41 = 64,$$

which gives $y = 4$. Also note that $(3, 4)$ satisfies the given equation. Hence, it is the only solution over the positive integers. Thus, the maximum possible value of $x + y$ is $3 + 4 = 7$. ■

Exercise 27 (IOQM 2023 P9,). Find the number of triples (a, b, c) of positive integers such that

(a) ab is a prime,

(b) bc is a product of two primes,

(c) abc is not divisible by square of any prime and

(d) $abc \leq 30$.

Walkthrough —

(a) Suppose (a, b, c) is a pair satisfying the above conditions.

(b) Since ab is a prime, it follows that one of a, b is equal to 1, and the remaining one is a prime.

(c) If $a = 1$, then b, c are distinct primes.

(d) If $b = 1$, then c is the product of two distinct primes which are not equal to a.

(e) This shows that (a, b, c) is equal to one of

$$(1, 2, 3), (1, 2, 5), (1, 2, 7), (1, 2, 11), (1, 2, 13),$$
$$(1, 3, 2), (1, 3, 5), (1, 3, 7),$$
$$(1, 5, 2), (1, 5, 3),$$
$$(1, 7, 2), (1, 7, 3),$$
$$(1, 11, 2),$$
$$(1, 13, 2),$$
$$(2, 1, 3 \cdot 5), (3, 1, 2 \cdot 5), (5, 1, 2 \cdot 3).$$

(f) Hence, the number of triples satisfying the above conditions is at most 17.

(g) Since any of the above triples does satisfy the given conditions, it follows that the number of triples satisfying the given conditions is equal to 17.

Solution 25. ■

Exercise 28 (Canadian Mathematical Olympiad 2019 P2,). Let a and b be positive integers such that $a + b^3$ is divisible by $a^2 + 3ab + 3b^2 - 1$. Prove that $a^2 + 3ab + 3b^2 - 1$ is divisible by the cube of an integer greater than 1.

Walkthrough —

(a) Show that $(a+b)^3$ is divisible by $a^2+3ab+3b^2-1$.

(b) Is it possible that $a^2+3ab+3b^2-1$ divides $(a+b)^2$?

Solution 26. Note that

$$\begin{aligned}(a+b)^3 &= a(a^2+3ab+3b^2)+b^3\\ &= a(a^2+3ab+3b^2-1)+a+b^3\end{aligned}$$

holds. Since $a+b^3$ is divisible by $a^2+3ab+3b^2-1$, it follows that $(a+b)^3$ is also divisible by $a^2+3ab+3b^2-1$. Assume for the sake of contradiction that $a^2+3ab+3b^2-1$ is not divisible by the cube of an integer greater than 1. It follows that $a^2+3ab+3b^2-1$ divides $(a+b)^2$. This implies that

$$a^2+2ab+b^2 \geq a^2+3ab+3b^2-1,$$

which simplifies to

$$ab+2b^2 \leq 1,$$

which is impossible since a and b are positive integers. This proves that $a^2+3ab+3b^2-1$ is divisible by the cube of an integer greater than 1. ■

Exercise 29 (British Mathematical Olympiad Round 1 2000 P2)**.** Show that, for every positive integer n, the integer

$$121^n-25^n+1900^n-(-4)^n$$

is divisible by 2000.

Walkthrough —

(a)

Solution 27. Let n be a positive integer. Note that

$$\begin{aligned}121^n-25^n+1900^n-(-4)^n &\equiv (-4)^n-25^n+25^n-(-4)^n \pmod{125}\\ &\equiv 0 \pmod{125}.\end{aligned}$$

Also note that

$$\begin{aligned}121^n-25^n+1900^n-(-4)^n &\equiv 9^n-9^n+(-20)^n-(-20)^n \pmod{16}\\ &\equiv 0 \pmod{16}.\end{aligned}$$

Since 125 and 16 are coprime, it follows that

$$121^n-25^n+1900^n-(-4)^n$$

is divisible by 2000 for all positive integers n. ■

Exercise 30 (British Mathematical Olympiad Round 1 2016/17 P6)**.** Consecutive positive integers $m, m+1, m+2, m+3$ are divisible by consecutive odd positive integers $n, n+2, n+4, n+6$ respectively. Determine the smallest possible value of m in terms of n.

Walkthrough —

(a)

Solution 28.

Claim — Let m be a positive integer and n be a positive odd integer such that the integers $m, m+1, m+2$ and $m+3$ are divisible by $n, n+2, n+4$ and $n+6$ respectively. Then m is at least as large as

$$\left(\frac{(n+4)(n+6)+1}{2}+\left(\frac{\mathfrak{n}-1}{2}\right)(n+4)(n+6)\right)(n+2)-1,$$

where

$$\mathfrak{n}=\begin{cases} n & \text{if 3 does not divide } n,\\ \frac{n}{3} & \text{if 3 divides } n.\end{cases}$$

Proof of the Claim. Let k, r, s, t be positive integers such that

$$m=kn, \quad m+1=r(n+2), \quad m+2=s(n+4), \quad m+3=t(n+6).$$

Using that n divides m, it follows that n divides $2r-1$.

Since $n+4$ divides $m+2$, it follows that $n+4$ divides $2r-1$. Finally, using that $n+6$ divides $m+3$, we obtain that $n+6$ divides $2r-1$.

Hence, the integer $2r-1$ is divisible by each of $n, n+4, n+6$. Since $n+4, n+6$ are consecutive odd integers, they are coprime. Thus, $(n+4)(n+6)$ divides $2r-1$. This shows that

$$r=\frac{(n+4)(n+6)+1}{2}+\alpha(n+4)(n+6)$$

for some non-negative integer α. Since n divides $2r-1$, it follows that n divides $24+48\alpha$. Using that n is odd, we obtain that n divides $3(2\alpha+1)$. Write

$$\mathfrak{n}=\begin{cases} n & \text{if 3 does not divide } n,\\ \frac{n}{3} & \text{if 3 divides } n.\end{cases}$$

Note that $\mathfrak{n}$ is odd and $\mathfrak{n}$ divides $2\alpha+1$. Thus, there exists a non-negative integer β such that

$$\alpha=\mathfrak{n}\beta+\frac{\mathfrak{n}-1}{2}.$$

The content posted here and at this blog by Evan Chen are quite useful.

Substituting this into the expression for r, we obtain

$$r = \frac{(n+4)(n+6)+1}{2} + \left(\mathfrak{n}\beta + \frac{\mathfrak{n}-1}{2}\right)(n+4)(n+6).$$

□

Claim — Let R denote the positive integer

$$\frac{(n+4)(n+6)+1}{2} + \left(\frac{\mathfrak{n}-1}{2}\right)(n+4)(n+6),$$

and let M denote the positive integer

$$(n+2)R - 1.$$

Then, the integers $M, M+1, M+2, M+3$ are divisible by $n, n+2, n+4, n+6$ respectively.

Proof of the Claim. Note that

$$\begin{aligned} 2R - 1 &\equiv 24 + 24(\mathfrak{n}-1) \pmod{n} \\ &\equiv 24\mathfrak{n} \pmod{n} \\ &\equiv 0 \pmod{n}. \end{aligned}$$

This shows that n divides M. Also note that $n+2$ divides $M+1$. Moreover,

$$\begin{aligned} &M+2 \\ &= (n+2)R + 1 \\ &= \frac{(n+2)(n+4)(n+6)+n+4}{2} + \left(\frac{\mathfrak{n}-1}{2}\right)(n+2)(n+4)(n+6) \\ &= \frac{(n+2)(n+6)+1}{2}(n+4) + \left(\frac{\mathfrak{n}-1}{2}\right)(n+2)(n+4)(n+6) \end{aligned}$$

holds, which shows that $n+4$ divides $M+2$. Similarly, it follows that

$$M+3 = \frac{(n+2)(n+4)+1}{2}(n+6) + \left(\frac{\mathfrak{n}-1}{2}\right)(n+2)(n+4)(n+6),$$

which shows that $n+6$ divides $M+3$. This completes the proof of the claim. □

Combining the two claims, it follows that the smallest possible value of m is equal to M, that is, the integer

$$\frac{(n+2)(n+4)(n+6)+n}{2} + \left(\frac{\mathfrak{n}-1}{2}\right)(n+2)(n+4)(n+6),$$

which is equal to

$$\frac{1}{2}\left(n + \mathfrak{n}(n+2)(n+4)(n+6)\right).$$

■

Exercise 31 (British Mathematical Olympiad Round 1 2019 P1)**.** Show that there are at least three prime numbers p less than 200 for which $p+2$, $p+6$, $p+8$ and $p+12$ are all prime. Show also that there is only one prime number q for which $q+2$, $q+6$, $q+8$, $q+12$ and $q+14$ are all prime.

Walkthrough —

(a) Let p be a prime number such that p, $p+2$, $p+6$, $p+8$ and $p+12$ are all prime.

(b) Show that p is odd.

(c) Using that $p+6, p+8$ are primes greater than 3, show that $p \equiv 2 \pmod 3$.

(d) Using that $p+6, p+8, p+12$ are primes greater than 5, show that $p = 5$ or $p \equiv 1 \pmod 5$.

(e) Using that $p+2, p+6, p+8, p+12$ are primes greater than 7, show that $p = 7$ or p is congruent to either 3 or 4 modulo 7.

(f) Using the above congruence conditions modulo 3, 5 and 7, show that if $p \leq 200$, then p is equal to one of the integers

$$11, 41, 71, 101, 131, 161, 191.$$

(g) Complete the first part of the problem.

(h) For the second part of the problem, show that if q is a prime number such that q, $q+2$, $q+6$, $q+8$, $q+12$ and $q+14$ are all prime, then q is at most 5.

Solution 29. Let p be a prime number such that p, $p+2$, $p+6$, $p+8$ and $p+12$ are all prime. Note that p is odd. Since $p+6, p+8$ are primes greater than 3, it follows that

$$p \not\equiv 0 \pmod 3, \quad p \not\equiv 1 \pmod 3,$$

which implies that $p \equiv 2 \pmod 3$. Since $p+6, p+8, p+12$ are primes greater than 5, it follows that

$$p \not\equiv 4 \pmod 5, \quad p \not\equiv 2 \pmod 5, \quad p \not\equiv 3 \pmod 5,$$

which implies that $p = 5$ or $p \equiv 1 \pmod 5$. Note that if $p = 5$, then $p+2, p+6, p+8, p+12$ are all prime. Henceforth, let us assume that $p \equiv 1 \pmod 5$. Note that $p+2, p+6, p+8, p+12$ are all primes greater than 7. It follows that p not congruent to any of $5, 1, 6, 2$ modulo 7. This shows that $p = 7$ or p

is congruent to either 3 or 4 modulo 7. Note that if $p = 7$, then $p + 2$ is not a prime. This implies that p is congruent to either 3 or 4 modulo 7. Since $p \equiv 2 \pmod 3$ and $p \equiv 1 \pmod 5$, it follows that p is congruent to 11 modulo 15. If $p \leq 200$, then p is equal to one of the integers

$$11, 41, 71, 101, 131, 161, 191.$$

Among the above integers, only $11, 101$ satisfy the required congruence condition modulo 7. Note that if $p = 11$, then $p + 2, p + 6, p + 8, p + 12$ are all prime. Moreover, if $p = 101$, then $p + 2, p + 6, p + 8, p + 12$ are all prime. This shows that there are precisely three prime numbers p less than 200 for which $p + 2, p + 6, p + 8, p + 12$ are all prime, and these are $5, 11, 101$.

Let q be a prime number such that $q, q + 2, q + 6, q + 8, q + 12, q + 14$ are all prime. Since $q+6, q+8$ are primes greater than 3, it follows that $q \equiv 2 \pmod 3$. Note that q is odd, and hence, q is at least 5. Since $q + 6, q + 8, q + 12, q + 14$ are primes greater than 5, it follows that q is equal to 5. Note that if $q = 5$, then $q+2, q+6, q+8, q+12, q+14$ are all prime. This shows that there is only one prime number q for which $q + 2, q + 6, q + 8, q + 12, q + 14$ are all prime, and this is 5. ■

Exercise 32 (British Mathematical Olympiad Round 1 2024 P5)**.** Let p be a prime number, and let n be the smallest positive integer, strictly greater than 1, for which $n^6 - 1$ is divisible by p. Prove that at least one of $(n+1)^6 - 1$ and $(n + 2)^6 - 1$ is divisible by p.

Walkthrough —

(a) Consider the cases that p divides $n^2 - n + 1$ and that p does not divide $n^2 - n + 1$.

(b) Observe that

$$n^2 - n + 1 = (n - 1)^2 + (n - 1) + 1,$$
$$n^2 + n + 1 = (n + 1)^2 - (n + 1) + 1$$

hold.

Solution 30. Let p be a prime number, and let n be the smallest positive integer, strictly greater than 1, for which $n^6 - 1$ is divisible by p.

Let us consider the case that p divides $n^2 - n + 1$. Note that

$$n^2 - n + 1 = (n - 1)^2 + (n - 1) + 1$$

holds. Then p divides $(n-1)^2+(n-1)+1$, implying that p divides $(n-1)^6-1$. Using the minimality of n, it follows that $n - 1$ is equal to 1. This implies that p divides $2^2 - 2 + 1 = 3$. This shows that p is equal to 3. Using $p = 3$ and $n = 2$, it follows that $(n + 2)^6 - 1$ is divisible by p.

Now, let us consider the case that p does not divide $n^2 - n + 1$. If $p = 2$, then p divides $(n + 2)^6 - 1$. Henceforth, let us assume that p is an odd prime. Since p divides $(p - 1)^6 - 1$, using the minimality of n and that $p - 1 \geq 2$, it follows that n is less than or equal to $p - 1$. If n is less than $p - 1$, then using the factorization

$$n^6 - 1 = (n - 1)(n + 1)(n^2 - n + 1)(n^2 + n + 1),$$

it follows that p divides $n^2 + n + 1$, and as consequence, p divides $(n + 1)^2 - (n + 1) + 1$, which is a divisor of $(n + 1)^6 - 1$. Finally, if n is equal to $p - 1$, then p divides $(n + 2)^6 - 1$. This completes the proof. ■

Example 33 (European Mathematical Cup Junior 2013 P1)**.** For a positive integer m, let $m?$ be the product of first m prime numbers. Determine if there exist positive integers m and n with the following property:

$$m? = n(n + 1)(n + 2)(n + 3).$$

> **Walkthrough** — The product of four consecutive positive integers is divisible by 4.

Exercise 34 (Tournament of Towns Spring 2019, Junior, O Level P4)**.** The product of two positive integers m and n is divisible by their sum. Prove that $m + n \leq n^2$.

> **Walkthrough** — Note that if $m + n$ divides mn, then $m + n$ divides $n(m + n) - mn$.

Example 35. Among any four consecutive positive integers, one of them is coprime to (that is, has no common factor with) the remaining three.

> **Walkthrough** — Show that among any four consecutive positive integers, at least one of the odd integers is not divisible by 3. Consider the case when this odd integer is equal to 1, and the case when it it greater than one. In the second case, find a suitable prime divisor of this odd integer.

Proof. Note that among any four consecutive positive integers, at least one of the odd integers is not divisible by 3, and hence, either it is equal to 1, in which case it is coprime to the remaining ones, or it is greater than one, and its smallest prime factor is at least 5, and hence, it is coprime to the remaining ones. □

Exercise 36 (Tournament of Towns Fall 2019, Junior, O Level P4, by Boris Frenkin)**.** There are given 1000 integers $a_1, \ldots, a_{1000}$. Their squares $a_1^2, \ldots, a_{1000}^2$ are written along the circumference of a circle. It so happened that the sum of any 41 consecutive numbers on this circle is a multiple of 41^2. Is it necessarily true that every integer $a_1, \ldots, a_{1000}$ is a multiple of 41?

Remark. Replace 1000 by 10 and 41 by 7, and try to work on the problem.

Walkthrough —

(a) Show that if $1 \leq i, j \leq 1000$ are integers satisfying $i \equiv j \bmod 41$, then $a_i^2 \equiv a_j^2 \bmod 41^2$.

(b) In particular, show that $a_1^2 \equiv a_{41k+1}^2 \bmod 41^2$ for any integer k.

(c) Does it follow that a_1^2 is congruent to a_i^2 modulo 41^2 for any integer $1 \leq i \leq 1000$?

(d) Show that $41a_1^2$ is congruent to $a_1^2 + a_2^2 + \cdots + a_{41}^2$ modulo 41^2, and conclude.

Solution 31. For any integer m, let $\overline{m}$ denote the integer lying between 1 and 1000, which is congruent to m modulo 1000. Note that

$$a_i^2 \equiv a_j^2 \bmod 41^2$$

holds for any integers i, j lying between 1 and 1000, and satisfying $i \equiv j \bmod 41$. It follows that

$$a_1^2 \equiv a_{41k+1}^2 \bmod 41^2$$

for any integer k. Since the integers $41, 1000$ are relatively prime, it follows that the integers

$$41, 41 \cdot 2, 41 \cdot 3, \ldots, 41 \cdot 1000$$

are pairwise distinct modulo 1000, that is, these integers are congruent to $1, 2, \ldots, 1000$ modulo 1000 in some order. This shows that a_1^2 is congruent to a_i^2 modulo 41^2 for any integer $1 \leq i \leq 1000$. It follows that

$$41a_1^2 \equiv a_1^2 + a_2^2 + \cdots + a_{41}^2 \bmod 41^2$$

Since the sum $a_1^2 + a_2^2 + \cdots + a_{41}^2$ is divisible by 41^2, this shows that 41 divides a_1. For any integer $1 \leq i \leq 1000$, 41^2 divides $a_1^2 - a_i^2$, and using that 41 divides a_1, we obtain 41 divides a_i.

This proves that it is necessary that every integer $a_1, \ldots, a_{1000}$ is a multiple of 41. ■

Example 37. [FGI96, Problem 83, p. 72] Prove that if a prime number is divided by 30, the remainder is a prime or 1.

Solution 32. Let p be a prime number. If p is less than 5, then we are done. Henceforth, let us assume that $p \geq 5$. It follows that p is of the form $6k \pm 1$ for some positive integer k. If $k \equiv 1 \pmod 5$, then p is equivalent to one of $5, 7$ modulo 30. If $k \equiv 2 \pmod 5$, then p is equivalent to one of $1, 3$ modulo 30. If $k \equiv 3 \pmod 5$, then p is equivalent to 2 modulo 30. If $k \equiv 4 \pmod 5$, then p is equivalent to one of $3, 5$ modulo 30. This completes the proof. ■

Exercise 38 (RMO 2004 P6, , Australian MO 1982)**.** Let $(p_1, p_2, p_3, \dots, p_n, \dots)$ be a sequence of primes, defined by $p_1 = 2$ and for $n \geq 1$, p_{n+1} is the largest prime factor of $p_1 p_2 \cdots p_n + 1$. Prove that $p_n \neq 5$ for any n.

Walkthrough —

(a) Show that $p_1 p_2 p_3 \cdots p_n + 1$ is odd for any $n \geq 1$, and p_n is odd for any $n \geq 2$. Deduce that $p_1 p_2 p_3 \dots p_n + 1$ is not a multiple of 3. (If you are stuck, then does verifying this statement for small values of n help?)

(b) What can be said about the smallest prime divisor of $p_1 p_2 p_3 \dots p_n + 1$?

(c) If it is a power of 5, then $p_1 p_2 p_3 \dots p_n$ is divisible by 4. Arrive at a contradiction.

Solution 33. Note that $p_1 p_2 \dots p_n + 1$ is odd for any $n \geq 1$, and hence p_n is odd for any $n \geq 2$. Since $p_1 = 2$ and $p_2 = 3$, it follows that for any $n \geq 2$, the integer $p_1 p_2 \dots p_n + 1$ is not divisible by any one of 2 and 3. So the least prime divisor of $p_1 p_2 \dots p_n + 1$ is at least 5 for any $n \geq 2$. If possible, suppose 5 is the largest prime divisor of $p_1 p_2 \dots p_n + 1$ for some integer $n \geq 2$. This yields

$$p_1 p_2 \dots p_n + 1 = 5^k$$

for some $k \geq 1$. This implies that 4 divides $p_1 p_2 \cdots p_n$, which is impossible since $p_1 = 2$, and p_r is odd for any integer $r \geq 2$. This shows that p_{n+1} is not equal to 5 for any integer $n \geq 2$. Consequently, it follows that $p_n \neq 5$ for any integer $n \geq 1$. ■

Exercise 39 (Moscow Mathematical Olympiad 1973 Grade 8 Second Round P4)**.** Prove that the equation

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{p},$$

where x, y are positive integers, has exactly 3 solutions if p is a prime and the number of solutions is greater than three if $p > 1$ is not a prime. We consider solutions (a, b) and (b, a) for $a \neq b$ distinct.

Example 40 (cf. Moscow MO 1973 Day 1 Grade 8 P4 **??**)**.** For any positive integer n, show that the number of ordered pairs (x, y) of positive integers for which

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

is equal to the number of positive divisors of n^2.

Solution 34. For positive integers x, y, note that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

holds if and only if

$$(x - n)(y - n) = n^2$$

holds. Observe that if x, y are positive integers satisfying the given equation, then $x > n$ and $y > n$ holds. This shows that the solutions of the given equation over the positive integers are in one-to-one correspondence with pairs of positive integers (a, b) such that $ab = n^2$, through the map

$$(a + n, b + n) \leftrightarrow (a, b).$$

This completes the proof. ■

Exercise 41 (INMO 1991 P10, , cf. Moscow Mathematical Olympiad 1973 Grade 8 First Round P4, UK BMO 2005 Round 2 P1)**.** For any positive integer n, let $S(n)$ denote the number of ordered pairs (x, y) of positive integers for which

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

(for instance, $S(2) = 3$). Determine the set of positive integers n for which $S(n) = 5$.

> **Walkthrough** — Is the given equation equivalent to
>
> $$(x - n)(y - n) = n^2?$$

Solution 35. For positive integers x, y, note that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

holds if and only if

$$(x - n)(y - n) = n^2$$

holds. Observe that if x, y are positive integers satisfying the given equation, then $x > n$ and $y > n$ holds. This shows that the solutions of the given equation over the positive integers are in one-to-one correspondence with the pairs of positive integers (a, b) such that $ab = n^2$, through the map

$$(a + n, b + n) \leftrightarrow (a, b).$$

Hence, the set of positive integers n satisfying $S(n) = 5$ is equal to the set of positive integers n such that n^2 has precisely 5 positive divisors. Note that any

such integer n is larger than 1. Writing n as a product of powers of distinct primes, it follows that n^2 has precisely 5 positive divisors if and only if n is the square of a prime. Indeed, if $p_1, \ldots, p_r$ are distinct primes, and $a_1, \ldots, a_r$ are positive integers, then the integer $(p_1^{a_1} \ldots p_r^{a_r})^2$ has precisely 5 positive divisors if and only if

$$(2a_1 + 1)(2a_2 + 1) \ldots (2a_r + 1) = 5$$

holds, which is equivalent to $r = 1, a_1 = 2$. This proves that the positive integers satisfying $S(n) = 5$ are precisely the squares of the primes. ■

Exercise 42 (RMO 1992 P2, , cf. Moscow Mathematical Olympiad 1973 Grade 8 First Round P4)**.** If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$, where a, b, c are positive integers with no common factor, prove that $(a + b)$ is the square of an integer.

Walkthrough — Is the given equation equivalent to

$$(a - c)(b - c) = c^2?$$

Solution 36. Let a, b, c be positive integers satisfying the given equation. Assume that a, b have no common prime factors. Note that $(a - c)(b - c) = c^2$ holds. Also note that any common prime divisor of $a - c, b - c$ divides $(a - c)(b - c) = c^2$, and hence it divides the integers a, b, which is impossible. This shows that the integers $a - c, b - c$ are relatively prime, and satisfy $(a - c)(b - c) = c^2$. Note also that $a > c$ holds. Hence, there exist relatively prime positive integers x, y such that $c = xy$, $a - c = x^2$ and $b - c = y^2$ holds. This gives

$$a = c + x^2 = xy + x^2, \quad b = c + y^2 = xy + y^2.$$

This implies that $a + b$ is a perfect square. ■

Example 43 (UK BMO 2005 Round 2 P1, cf. Moscow Mathematical Olympiad 1973 Grade 8 First Round P4, Exercise 39)**.** The integer N is positive. There are exactly 2005 ordered pairs (x, y) of positive integers satisfying

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{N}.$$

Show that N is a perfect square.

Walkthrough —

(a) Note that it suffices to show that if N^2 has precisely 2005 positive divisors for some positive integer N, then N is a perfect square.

(b) Note that $2005 = 5 \cdot 401$.

(c) Observe that 401 is a prime, and hence, all the prime factors of 2005 are

congruent to 1 modulo 4.

Example 44 (All-Russian MO 1989 Day 2 Grade 8 P5)**.** Show that number $4^{545} + 545^4$ is composite.

Exercise 45 (RMO 1991 P7, , cf. All-Russian MO 1989 Grade 8 Day 2 P5 Example 44, India BStat 2006 P3)**.** [**Eng98**, p. 121] Show that $n^4 + 4^n$ is prime if and only if $n = 1$.

Solution 37. Let n be a positive integer such that $n^4 + 4^n$ is a prime. Note that n is odd, and

$$\begin{aligned} n^4 + 4^n &= (n^2 + 2^n)^2 - 2^{n+1}n^2 \\ &= (n^2 + 2^n)^2 - (2^{(n+1)/2}n)^2 \\ &= (n^2 + 2^n - 2^{(n+1)/2}n)(n^2 + 2^n + 2^{(n+1)/2}n) \end{aligned}$$

holds. Since $n^4 + 4^n, n^2 + 2^n + 2^{(n+1)/2}n$ are positive, it follows that $n^2 + 2^n - 2^{(n+1)/2}n$ is positive. This gives $n^2 + 2^n - 2^{(n+1)/2}n = 1$. Note that

$$n^2 + 2^n - 2^{(n+1)/2}n \geq 2 \cdot 2^{n/2}n - 2^{(n+1)/2}n = (\sqrt{2} - 1)2^{(n+1)/2}n$$

holds. If $n \geq 2$, then the above yields

$$n^2 + 2^n - 2^{(n+1)/2}n \geq (\sqrt{2} - 1)2^{(n+1)/2}n > 1,$$

which implies that $n^4 + 4^n$ is not a prime. This proves that $n = 1$. Moreover, if $n = 1$, then $n^4 + 4^n$ is a prime. This completes the proof. ■

Example 46 (India INMO 1991 P1)**.** Find the number of positive integers n for which

1. $n \leq 1991$ and
2. 6 is a factor of $(n^2 + 3n + 2)$.

Solution 38. Note that if n is a positive integer, then 6 divides $(n+1)(n+2)$ if and only if 3 divides $(n+1)(n+2)$, which is equivalent to $n \equiv \pm 1 \bmod 3$. So, the positive integers satisfying the given conditions are precisely the positive integers at most 1991, which are not divisible by 3. Noting that $1991 = 3 \cdot 663 + 2$, it follows that the requires number is equal to

$$664 + 664 = 1328.$$

■

Exercise 47 (RMO 1991 P3,). A four-digit number has the following properties:

1. it is a perfect square,
2. its first two digits are equal to each other,
3. its last two digits are equal to each other.

Find all such four-digit numbers.

Solution 39. Let n be a positive integer satisfying the given conditions. Denote the first two digits of n by a, and the last two digits of n by b. Since n is a perfect square, it follows that n is congruent to one of $0, 1$ modulo 4, which shows that $10b + b = 11b$ is congruent to one of $0, 1$ modulo 4, or equivalently, b is congruent to one of $0, 3$ modulo 4. Similarly, using that a perfect square is congruent to one of $0, 1, -1$ modulo 5, it follows that b is congruent to one of $-1, 0, 1$ modulo 5. This implies that b is equal to one of $0, 4$.

If $b = 0$, then n is the square of a multiple of 10, and hence, its first two digits would not be equal. It follows that $b = 4$, and consequently, we obtain

$$n = 1000a + 100a + 10b + b = 1100a + 11b = 1100a + 44.$$

This shows that 11 divides the perfect square n. So, 11^2 divides n, and hence, 11 divides $100a + 4$, which yields

$$a \equiv 7 \bmod 11.$$

Since $1 \leq a \leq 9$, we obtain $a = 7$, and this gives $n = 7744$. Note that $7744 = 88^2$, it follows that 7744 is the only four-digit number satisfying the given conditions. ■

Exercise 48 (RMO 2011b P2, , All-Russian MO 1992–1993 Grade 9 Final Stage P1). Let n be a positive integer such that $2n + 1$ and $3n + 1$ are both perfect squares. Show that $5n + 3$ is a composite number.

Solution 40. Let a, b be positive integers such that

$$2n + 1 = a^2, \quad 3n + 1 = b^2$$

holds. Note that

$$5n + 3 = 4(2n + 1) - (3n + 1) = 4a^2 - b^2 = (2a - b)(2a + b). \tag{3}$$

Since $2a + b \geq 2$, it follows that $2a - b$ is positive, and hence, it suffices to show that $2a - b \neq 1$. Using $2n + 1$ is a perfect square, it follows that $n \geq 3$. Note that

$$2a - b = 2\sqrt{2n + 1} - \sqrt{3n + 1}$$

$$
\begin{aligned}
&= \frac{4(2n+1)-(3n+1)}{2\sqrt{2n+1}+\sqrt{3n+1}} \\
&= \frac{5n}{2\sqrt{2n+1}+\sqrt{3n+1}} \\
&\geq \frac{5n}{2\sqrt{3n}+\sqrt{4n}} \\
&= \frac{5}{2+2\sqrt{3}}\sqrt{n} \\
&> 1,
\end{aligned}
$$

where the last inequality is obtained using $n \geq 3$. This shows that $2a - b > 1$. From Eq. (3), it follows that $5n + 3$ is a composite number. ■

Exercise 49 (RMO 1994 P5,). Let A be a set of 16 positive integers with the property that the product of any two distinct numbers of A will not exceed 1994. Show that there are two numbers a and b in A which are not relatively prime.

Walkthrough —

(a) Suppose $\mathcal{A}$ is a set of 16 positive integers, and assume that $\mathcal{A}$ does not satisfy the conclusion of the problem, that is, it is false that **some two** *numbers in* $\mathcal{A}$ **are not** *relatively prime.* In other words, $\mathcal{A}$ has the property that **any two** *numbers in* $\mathcal{A}$ **are** *relatively prime.*

(b) Prove that such a set $\mathcal{A}$ would fail to satisfy the given condition, which states that *the product of* **any two** *distinct elements of* $\mathcal{A}$ *is smaller than* 1994. In other words, show that *the product of* **some two** *elements of* $\mathcal{A}$ *is greater than or equal to* 1994.

Remark. Observe that the above argument shows that if a set of 16 positive integers violates the conclusion of the problem, then it does not satisfy the given condition. **Convince yourself** that **doing so does prove** that if a set of 16 positive integers satisfy the given condition, then **there is no way that** it would fail to satisfy the conclusion.

Further, also **convince yourself** that establishing that the failure of the conclusion forces the failure of the given condition [a] is **equivalent to** establishing that the given condition implies the stated conclusion.

[a] When there are more than one condition, the given conditions are to be **considered together**, and the *failure of the* **totality** *of the given conditions* means the *failure of* **at least one of** *the given conditions.*

Solution 41. Note that if $n \geq 1$, and $a_1, \ldots, a_n$ are distinct and pairwise coprime positive integers such that $a_i \geq 2$ for all i, then one of them admits a prime factor which is at least as large as the n-th prime. Indeed, for each i, if

we fix a prime divisor p_i of a_i, then using that $a_1, \ldots, a_n$ are pairwise coprime, it follows that $p_1, \ldots, p_n$ are distinct primes, and hence the largest among them is at least as large as the n-th prime. Consequently, if $n \geq 2$, and $a_1, \ldots, a_n$ are distinct and pairwise coprime positive integers, then at least $(n-1)$ of them are greater than 1, and hence one of them is divisible by a prime at least as large as the $(n-1)$-st prime.

If possible, let us assume that the elements of A are pairwise coprime. Hence, A contains an element x which is divisible by a prime at least as large as the 15th prime. Similarly, the set $A \setminus \{x\}$ contains an element y which is divisible by a prime at least as large as the 14th prime. Since the 14th and 15th primes are $43, 47$ respectively, it follows that $x \geq 47, y \geq 43$, and consequently,

$$xy \geq 47 \cdot 43 = 2021 > 1994,$$

which contradicts the hypothesis. This proves that there are two numbers a and b in A which are not relatively prime. ■

Remark. Note that the above problem is similar to the following problem in spirit.

Example 50. For a set A of consisting of positive integers, let $\ell(A)$ denote the largest integer which can be expressed as the product of two distinct elements of A. What is the smallest element of the set which consists of the integers of the form $\ell(A)$ as A runs over the sets of size 16 and consisting of positive integers?

Bogus Solution. Since $\ell(A)$ is to be minimized as A runs over the sets 16 pairwise coprime integers and $\ell(A)$ denotes the maximum of the products of the pairs of elements of A, it follows that the minimum value of $\ell(A)$ is achieved precisely when the elements of A are as small as possible. This shows that the minimum value of $\ell(A)$ occurs when

$$A = \{1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}.$$

Hence, the minimum value of $\ell(A)$ is $43 * 47 = 2021$.

Exercise 51. What goes wrong with the above?

Exercise 52 (RMO 1994 P3,). Find all 6-digit natural numbers $a_1a_2a_3a_4a_5a_6$ formed by using the digits $1, 2, 3, 4, 5, 6$, once each such that the number $a_1a_2 \ldots a_k$ is divisible by k, for $1 \leq k \leq 6$.

Walkthrough —

(a) Show that a_2, a_4, a_6 are equal to $2, 4, 6$ in some order.

(b) Prove that $a_5 = 5$, and a_1, a_3 are equal to $1, 3$ in some order.

(c) Using that 3 divides $a_1a_2a_3$, determine a_2.

(d) Using the divisibility condition by 4, show that $a_4 = 6$, and conclude that $a_6 = 4$.

Solution 42. Since $a_1a_2, a_1a_2a_3a_4, a_1a_2a_3a_4a_5a_6$ are divisible by 2, it follows that a_2, a_4, a_6 are even, and hence they are equal to $2, 4, 6$ in some order. Using that $a_1a_2a_3a_4a_5$ is divisible by 5, we get that $a_5 = 5$. So a_1, a_3 are equal to $1, 3$ in some order. Using that $a_1a_2a_3$ is a multiple of 3, we obtain

$$a_1 + a_2 + a_3 \equiv 0 \bmod 3,$$

which yields

$$a_2 \equiv 2 \bmod 3,$$

and hence, $a_2 = 2$ holds. Note that $1234, 3214$ are not divisible by 4. This shows that $a_4 = 6$, and hence $a_6 = 4$. It follows that $a_1a_2a_3a_4a_5a_6$ is equal to 321654, or 123654. Note that the integers $321654, 123654$ satisfy the required conditions too. This proves that $321654, 123654$ are precisely all the 6-digit numbers satisfying the given condition. ■

Exercise 53 (RMO 1995 P3, , cf. Tournament of Towns)**.** [**Tao06**, Problem 2.1] Prove that among any 18 consecutive three digit numbers there is at least one number which is divisible by the sum of its digits.

Walkthrough —

(a) Show that one among any such consecutive integers is divisible by 18.

(b) Prove that its sum of digits, is a multiple of 9, and conclude that it is equal to one of $9, 18, 27$.

(c) Show that the sum of its digits is not 27.

Solution 43. Note that among 18 consecutive three digit numbers, there is an integer divisible by 18. Denote it by $n = 100a + 10b + c$ with a, b, c denoting integers lying between 0 and 9. It follows that 9 divides n, and hence 9 divides $a + b + c$. This shows that $a + b + c$ is equal to one of $9, 18, 27$. Note that $a + b + c = 27$ holds only if $n = 999$. Since 18 divides n, it follows that $a + b + c \neq 27$, and hence, $a + b + c$ is equal to one of $9, 18$. This proves that $a + b + c$ divides n. ■

Exercise 54 (China TST 1995 Day 1 P1,)**.** Find the smallest prime number p that cannot be represented in the form $|3^a - 2^b|$, where a and b are non-negative integers.

Walkthrough —

(a) Any prime smaller than 41 can be expressed as the absolute value of the difference of a nonnegative power of 3 and a nonnegative power of 2.

(b) If $41 = 2^b - 3^a$, then $b \geq 3$ and hence $3^a \equiv -1 \bmod 8$, which is impossible.

(c) Assume that $41 = 3^a - 2^b$. Considering congruence modulo 3, show that b is an even positive integer. Reduce modulo 4 to show that a is even.

(d) Write $a = 2x, b = 2y$, and factorize 41.

(e) Conclude by obtaining a contradiction.

Solution 44. Note that any prime smaller than 41 can be expressed as the absolute value of the difference of a nonnegative power of 3 and a nonnegative power of 2, as shown below.

$$\begin{aligned} 2 &= 3 - 1, \\ 3 &= 4 - 1, \\ 5 &= 9 - 4, \\ 7 &= 8 - 1, \\ 11 &= 27 - 16, \\ 13 &= 16 - 3, \\ 17 &= 81 - 64, \\ 19 &= 27 - 8, \\ 23 &= 32 - 9, \\ 29 &= 32 - 3, \\ 31 &= 32 - 1, \\ 37 &= 64 - 27. \end{aligned}$$

Let us prove the following claim.

Claim — The prime number 41 cannot be expressed as the absolute value of the difference of a nonnegative power of 3 and a nonnegative power of 2.

Proof of the Claim. On the contrary, let us assume that

$$41 = |3^a - 2^b|$$

holds for some nonnegative integers a, b.

First, let us consider the case that $41 = 2^b - 3^a$. Note that $b \geq 3$ holds, and reducing the above modulo 8, it follows that $3^a \equiv -1 \pmod 8$, which is impossible.

Now, let us consider the case that $41 = 3^a - 2^b$. Reducing modulo 3, it follows that $2^b \equiv 1 \pmod 3$, which shows that b is even. Note that b is nonzero. Next, reducing modulo 4, we obtain $3^a \equiv 1 \pmod 4$, which implies that a is even. Writing $a = 2x, b = 2y$ for some positive integers x, y, we obtain

$$41 = 3^{2x} - 2^{2y} = (3^x - 2^y)(3^x + 2^y)$$

with $1 \leq 3^x - 2^y < 3^x + 2^y$, which yields

$$3^x - 2^y = 1, 3^x + 2^y = 41,$$

which is impossible.

Considering the above cases, the claim follows. □

This proves that 41 is smallest prime that cannot be expressed in the given form. ■

Exercise 55 (RMO 1995 P2,). Call a positive integer n *good* if there are n integers, positive or negative, and not necessarily distinct, such that their sum and products are both equal to n. Show that the integers of the form $4k + 1$ and 4ℓ are good.

Solution 45. Note that 1 is good. Let k be a positive integer. Put

$$a_1 = \cdots = a_k = 1, \quad b_1 = \cdots = b_k = -1.$$

Noting that

$$4k + 1 = (4k + 1) + 2(a_1 + \cdots + a_k) + 2(b_1 + \cdots + b_k),$$
$$4k + 1 = (4k + 1)\left(\prod_{i=1}^{k} a_i b_i\right)^2,$$

it follows that $4k+1$ is good. Also note that for a positive integer ℓ, 4ℓ is equal to the sum, and also equal to the product, of $2, 2, \ell$ and $4\ell - 2 - 2 - \ell$ many 1's, and hence, 4ℓ is good. ■

Exercise 56 (RMO 1996 P4,). Suppose N is an n-digit positive integer such that

1. all the n-digits are distinct, and
2. the sum of any three consecutive digits is divisible by 5.

Prove that n is at most 6. Further, show that starting with any digit one can find a six-digit number with these properties.

Solution 46. If $n \geq 7$, then note that the first digit of N is congruent to its fourth digit modulo 5, and its fourth digit is congruent to its seventh digit modulo 5, and hence, its first, fourth and seventh digits are congruent to each other modulo 5, which is impossible since the digits of n are distinct and no three integers among $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$ are congruent modulo 5. This shows that n is at most 6.

Let $0 \leq a \leq 4$ be an integer. Considering the 3-digit numbers

$$140, 230, 320, 410,$$

it follows that there are integers $0 \leq b, c \leq 4$ such that a, b, c are distinct, and their sum is divisible by 5. Taking

$$a_1 = a, a_2 = b, a_3 = c, a_4 = a_1 + 5, a_5 = a_2 + 5, a_6 = a_3 + 5,$$

it follows that the 6-digit integer

$$10^5 a_1 + 10^4 a_2 + 10^3 a_3 + 10^2 a_4 + 10 a_5 + a_6$$

has the required properties. ■

Exercise 57 (RMO 1997 P2,). For each positive integer n, define $a_n = 20 + n^2$, and $d_n = \gcd(a_n, a_{n+1})$. Find the set of all values that are taken by d_n and show by examples that each of these values are attained.

Solution 47. Let n be a positive integer. Since d_n divides the integers $20 + n^2, 20 + (n+1)^2$, it follows that d_n divides

$$20 + (n+1)^2 - (20 + n^2) = 2n + 1,$$

and d_n also divides

$$20 + (n+1)^2 + 20 + n^2 = 41 + 2n^2 + 2n.$$

This shows that d_n divides

$$2(41 + 2n^2 + 2n) - (2n+1)^2 = 81.$$

So, d_n is equal to one of $1, 3, 9, 27, 81$.

Note that

$$\begin{aligned} d_1 &= \gcd(a_1, a_2) \\ &= \gcd(21, 24) \\ &= 3, \\ d_2 &= \gcd(a_2, a_3) \\ &= \gcd(24, 29) \end{aligned}$$

$$= 1$$

holds.

If $d_n = 9$, then 9 divides $2n + 1$, showing that $n \equiv 4 \bmod 9$. Note that

$$d_4 = \gcd(a_4, a_5) = \gcd(36, 45) = 9$$

holds. If $d_n = 27$, then 27 divides $2n + 1$, implying that $n \equiv 13 \bmod 27$. Note that

$$d_{13} = \gcd(a_{13}, a_{14}) = \gcd(189, 216) = \gcd(189, 216 - 189) = \gcd(189, 27) = 27$$

holds. Also note that if $d_n = 81$, then 81 divides $2n + 1$, which shows that $n \equiv 40 \bmod 81$. Observe that

$$\begin{aligned} d_{40} &= \gcd(a_{40}, a_{41}) \\ &= \gcd(20 + 40^2, 20 + 41^2) \\ &= \gcd(1620, 2 \cdot 40 + 1) \\ &= 81 \end{aligned}$$

holds.

This proves that the set of values taken by d_n are precisely

$$1, 3, 9, 27, 81.$$

■

Exercise 58 (RMO 1998 P5,). Find the minimum possible least common multiple (lcm) of twenty (not necessarily distinct) natural numbers whose sum is 801.

Solution 48. Let L denote the minimum possible least common multiple of twenty (not necessarily distinct) natural numbers, whose sum is 801. Note that one of these natural numbers is larger than 40 (otherwise their sum would be no larger than $40 \times 20 = 800$), and so L is at least 41. Since 801 can be written as

$$3 + \underbrace{42 + 42 + \cdots + 42}_{19 \text{ times}} = 801,$$

we get $L \leq 42$. Let us prove the claim below.

Claim — The integer L is not equal to 41.

Proof of the Claim. On the contrary, assume that there exist twenty natural number numbers $x_1, \ldots, x_{20}$ satisfying

$$x_1 + \cdots + x_{20} = 801, \quad \operatorname{lcm}(x_1, \ldots, x_{20}) = 41.$$

Since 41 is a prime, it follows that 41 divides x_i for some $1 \leq i \leq 20$. Since 41 is the least common multiple of $x_1, \ldots, x_{20}$, we obtain $x_i = 41$. Note that the sum $x_1 + \cdots + x_{20}$ is less than $20 \cdot 41$. This shows that not all of $x_1 + \cdots + x_{20}$ is divisible by 41. Reordering the integers $x_1 + \cdots + x_{20}$ (if necessary), let $1 \leq k < 20$ denote the integer such that none of $x_1, x_2, \ldots, x_k$ is divisible by 41, and $x_{k+1}, x_{k+2}, \ldots, x_{20}$ are equal to 41. We obtain that

$$41 = \operatorname{lcm}(x_1, \ldots, x_{41}) = \operatorname{lcm}(\operatorname{lcm}(x_1, \ldots, x_k), 41).$$

Since none of $x_1, \ldots, x_k$ is divisible by 41, it follows that $\operatorname{lcm}(x_1, \ldots, x_k) = 1$, and hence,

$$x_1 = x_2 = \cdots = x_k = 1.$$

This yields

$$k + 41(20 - k) = 801,$$

which implies that $k \equiv 22 \bmod 41$, which is impossible since $1 \leq k \leq 19$. This proves the Claim. □

We obtain that $L = 42$. ■

Exercise 59 (RMO 1998 P2,). Let n be a positive integer and $p_1, p_2, \ldots, p_n$ be n prime numbers all larger than 5 such that 6 divides $p_1^2 + p_2^2 + \cdots + p_n^2$. Prove that 6 divides n.

> **Walkthrough** — Observe that any prime larger than 5 is congruent to ± 1 modulo 6.

Solution 49. Note that any prime number larger than 5 is of the form $6k \pm 1$. This yields

$$p_1^2 + p_2^2 + \cdots + p_n^2 \equiv (\pm 1)^2 + (\pm 1)^2 + \cdots + (\pm 1)^2 \bmod 6 \equiv n \bmod 6.$$

Since 6 divides $p_1^2 + p_2^2 + \cdots + p_n^2$, it follows that 6 divides n. ■

Exercise 60 (RMO 1999 P2,). Find the number of positive integers which divide 10^{999} but not 10^{998}.

Solution 50. Since every divisor of 10^{998} divides 10^{999}, the required number is equal to the difference of the number of divisors of 10^{999} and 10^{998}, which is equal to $(999 + 1)(999 + 1) - (998 + 1)(998 + 1) = 1999$. ■

Exercise 61 (Bay Area MO 1999 P1). Prove that among any 12 consecutive positive integers, there is at least one which is smaller than the sum of its proper divisors. (The proper divisors of a positive integer n are all positive integers other than 1 and n which divide n. For example, the proper divisors of 14 are 2 and 7.)

Walkthrough — $3, 4, \ldots$!

Solution 51. Among any twelve consecutive integers, there is a multiple of 12. For any positive integer n, note that

$$3n, 4n, 6n$$

are proper divisors of $12n$, and

$$12n < 3n + 4n + 6n$$

holds. This completes the proof. ■

Exercise 62 (Junior Balkan MO 2006 P3,). We call a number *perfect* if the sum of its positive integer divisors (including 1 and n) equals $2n$. Determine all perfect numbers n for which $n - 1$ and $n + 1$ are prime numbers.

Walkthrough —

(a) Note that 6 is a such a perfect number.

(b) Assume that $n > 6$ is a perfect number such that $n - 1$ and $n + 1$ are prime numbers.

(c) Show that n is divisible by 6.

(d) Prove that the sum of the positive integer divisors of n is greater than $2n$.

Solution 52. Note that 6 is a such a perfect number. Let $n > 6$ be a perfect number such that $n - 1$ and $n + 1$ are prime numbers. Since $n - 1$ is a prime number and $n - 1 \geq 3$, it follows that n is even. Also note that the integers $n - 1$ and $n + 1$ are larger than 3. Since these are primes, they are not divisible by 3. This shows that n is divisible by 3. Thus, n is divisible by 6. Write $n = 6m$ for some positive integer $m \geq 2$. Then, the sum of the positive integer divisors of n is

$$\geq 1 + m + 2m + 3m + 6m > 12m = 2n,$$

a contradiction. This proves that 6 is the only perfect number such that $n - 1$ and $n + 1$ are prime numbers. ■

Exercise 63 (Bay Area MO 2000 P1). Prove that any integer greater than or equal to 7 can be written as a sum of two relatively prime integers, both greater than 1. (Two integers are relatively prime if they share no common positive divisor other than 1. For example, 22 and 15 are relatively prime, and thus $37 = 22 + 15$ represents the number 37 in the desired way.)

> **Walkthrough** — Consider the case of an odd integer, the case of a multiple of 4, and the case of an even integer, which is not a multiple of 4.

Solution 53. Note that any odd integer can be expressed as the sum of two relatively prime integers. Indeed, for any integer n, the integer $2n+1$ is the sum of the relatively prime integers $n, n+1$.

For any integer k, note that

$$4k = (2k-1) + (2k+1)$$

holds, and the integers $2k-1, 2k+1$ are relatively prime since any of their common divisors is odd and divides $(2k+1)-(2k-1) = 2$.

For any integer ℓ, note that

$$4\ell + 2 = (2\ell - 1) + (2\ell + 3)$$

holds, and the integers $2\ell - 1, 2\ell + 3$ are relatively prime since any of their common divisors is odd and divides $(2\ell+3)-(2\ell-1) = 4$. ■

Exercise 64 (RMO 2000 P6,).

1. Consider two positive integers a and b which are such that $a^a b^b$ is divisible by 2000. What is the least possible value of the product ab?

2. Consider two positive integers a and b which are such that $a^b b^a$ is divisible by 2000. What is the least possible value of the product ab?

Solution 54. Let a, b denote positive integers such that $a^a b^b$ is divisible by 2000 and the product ab is minimum. Note that $2, 5$ divide ab, which shows that 10 divides ab, and hence $ab \geq 10$. Using that 2000 divides $10^{10}1^1$, we obtain $ab = 10$.

Now let a, b denote positive integers such that $a^b b^a$ is divisible by 2000 and the product ab is minimum. Note that $2, 5$ divide ab, and hence 10 divides ab. Moreover, $ab \leq 20$ since 2000 divides $4^5 5^4$. If $ab = 10$, then (a, b) is equal to one of

$$(1, 10), (2, 5), (5, 2), (10, 1),$$

and in each of this cases, 2000 does not divide $a^b b^a$. This shows that $ab \neq 10$, and $ab \leq 20$, and ab is a multiple of 10. Observing that 2000 divides $4^5 5^4$, we conclude that $ab = 20$. ■

Exercise 65 (RMO 2001 P7,). Prove that the product of the first 1000 positive even integers differs from the product of the first 1000 odd integers by a multiple of 2001.

Solution 55. Note that

$$\begin{aligned}&2\cdot 4\cdot 6\cdot \because 1998\cdot 2000\\&\quad -1\cdot 3\cdot 5\cdot \cdots \cdot 1997\cdot 1999\\&=(2001-1)(2001-3)\cdots(2001-1997)(2001-1999)\\&\quad -(2001-2)(2001-4)\cdots(2001-1998)(2001-2000)\end{aligned}$$

holds, which yields

$$\begin{aligned}&2\cdot 4\cdot 6\cdots 1998\cdot 2000-1\cdot 3\cdot 5\cdots 1997\cdot 1999\\&\equiv 1\cdot 3\cdot 5\cdots 1997\cdot 1999-2\cdot 4\cdot 6\cdots 1998\cdot 2000 \bmod 2001.\end{aligned}$$

It follows that

$$2(2\cdot 4\cdot 6\cdots 1998\cdot 2000-1\cdot 3\cdot 5\cdots 1997\cdot 1999)\equiv 0 \bmod 2001,$$

which implies that

$$2000(2\cdot 4\cdot 6\cdots 1998\cdot 2000-1\cdot 3\cdot 5\cdots 1997\cdot 1999)\equiv 0 \bmod 2001.$$

This shows that

$$2\cdot 4\cdot 6\cdots 1998\cdot 2000-1\cdot 3\cdot 5\cdots 1997\cdot 1999\equiv 0 \bmod 2001,$$

completing the proof. ■

Exercise 66 (RMO 2002 P3, , cf. RMO 2012d P2, RMO 2012a P2, RMO 2012b P2, RMO 2012c P2)**.** Let a,b,c be positive integers such that a divides b^2, b divides c^2 and c divides a^2. Prove that abc divides $(a+b+c)^7$.

Solution 56. In the following, k denotes one of the integers $2,3,4,5$. Assume that $a\mid b^k, b\mid c^k, c\mid a^k$. Note that if one of a,b,c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a,b,c are larger than 1. The given conditions imply that the set of prime divisors of a,b,c are equal.

Let p be a common prime divisor of a,b,c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a,b,c respectively. The given divisibility conditions imply that

$$\alpha\le k\beta, \beta\le k\gamma, \gamma\le k\alpha.$$

This yields

$$\alpha+\beta+\gamma\le \alpha+k^2\alpha+2\alpha=(k^2+k+1)\alpha,$$

and similarly, $\alpha+\beta+\gamma\le(k^2+k+1)\beta$, $\alpha+\beta+\gamma\le(k^2+k+1)\gamma$, which shows that

$$\alpha+\beta+\gamma\le(k^2+k+1)\min\{\alpha,\beta,\gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a+b+c)^{k^2+k+1}$. So abc divides $(a+b+c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a, b, c, the highest power of p that divides abc also divides $(a+b+c)^{k^2+k+1}$. It follows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Solution 57. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α, β, γ such that $\alpha+\beta+\gamma = k^2+k+1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α, β, γ are all positive. It remains to consider the case when exactly one of α, β, γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \geq k+1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \leq k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \geq k+1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \leq k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \geq k+1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \leq k$$

holds. This shows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Exercise 67 (RMO 2012d P2, , cf. RMO 2002 P3, RMO 2012a P2, RMO 2012b P2, RMO 2012c P2)**.** Let a, b, c be positive integers such that a divides b^2, b divides c^2 and c divides a^2. Prove that abc divides $(a+b+c)^7$.

Solution 58. In the following, k denotes one of the integers $2, 3, 4, 5$. Assume that $a \mid b^k, b \mid c^k, c \mid a^k$. Note that if one of a, b, c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a, b, c are larger than 1. The given conditions imply that the set of prime divisors of a, b, c are equal.

Let p be a common prime divisor of a, b, c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a, b, c respectively. The given divisibility conditions imply that

$$\alpha \leq k\beta, \beta \leq k\gamma, \gamma \leq k\alpha.$$

This yields

$$\alpha+\beta+\gamma \leq \alpha + k^2\alpha + 2\alpha = (k^2+k+1)\alpha,$$

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and similarly, $\alpha+\beta+\gamma \le (k^2+k+1)\beta$, $\alpha+\beta+\gamma \le (k^2+k+1)\gamma$, which shows that

$$\alpha+\beta+\gamma \le (k^2+k+1)\min\{\alpha,\beta,\gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a+b+c)^{k^2+k+1}$. So abc divides $(a+b+c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a,b,c, the highest power of p that divides abc also divides $(a+b+c)^{k^2+k+1}$. It follows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Solution 59. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α,β,γ such that $\alpha+\beta+\gamma = k^2+k+1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α,β,γ are all positive. It remains to consider the case when exactly one of α,β,γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \ge k+1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \le k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \ge k+1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \le k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \ge k+1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \le k$$

holds. This shows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Exercise 68 (RMO 2012a P2, cf. RMO 2002 P3, RMO 2012d P2, RMO 2012b P2, RMO 2012c P2)**.** Let a,b,c be positive integers such that $a|b^3, b|c^3$ and $c|a^3$. Prove that abc divides $(a+b+c)^{13}$.

Solution 60. In the following, k denotes one of the integers $2,3,4,5$. Assume that $a \mid b^k, b \mid c^k, c \mid a^k$. Note that if one of a,b,c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a,b,c are larger than 1. The given conditions imply that the set of prime divisors of a,b,c are equal.

Let p be a common prime divisor of a,b,c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a,b,c respectively. The given divisibility conditions imply that

$$\alpha \le k\beta, \beta \le k\gamma, \gamma \le k\alpha.$$

This yields

$$\alpha+\beta+\gamma \le \alpha + k^2\alpha + 2\alpha = (k^2+k+1)\alpha,$$

and similarly, $\alpha+\beta+\gamma \le (k^2+k+1)\beta$, $\alpha+\beta+\gamma \le (k^2+k+1)\gamma$, which shows that

$$\alpha+\beta+\gamma \le (k^2+k+1)\min\{\alpha,\beta,\gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a+b+c)^{k^2+k+1}$. So abc divides $(a+b+c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a,b,c, the highest power of p that divides abc also divides $(a+b+c)^{k^2+k+1}$. It follows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Solution 61. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α,β,γ such that $\alpha+\beta+\gamma = k^2+k+1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α,β,γ are all positive. It remains to consider the case when exactly one of α,β,γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \ge k+1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \le k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \ge k+1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \le k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \ge k+1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \le k$$

holds. This shows that abc divides $(a+b+c)^{k^2+k+1}$. ■

Exercise 69 (RMO 2012b P2, , cf. RMO 2002 P3, RMO 2012d P2, RMO 2012a P2, RMO 2012c P2)**.** Let a,b,c be positive integers such that $a|b^4, b|c^4$ and $c|a^4$. Prove that abc divides $(a+b+c)^{21}$.

Solution 62. In the following, k denotes one of the integers $2,3,4,5$. Assume that $a \mid b^k, b \mid c^k, c \mid a^k$. Note that if one of a,b,c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a,b,c are larger than 1. The given conditions imply that the set of prime divisors of a,b,c are equal.

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Let p be a common prime divisor of a, b, c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a, b, c respectively. The given divisibility conditions imply that

$$\alpha \le k\beta, \beta \le k\gamma, \gamma \le k\alpha.$$

This yields

$$\alpha + \beta + \gamma \le \alpha + k^2\alpha + 2\alpha = (k^2 + k + 1)\alpha,$$

and similarly, $\alpha + \beta + \gamma \le (k^2 + k + 1)\beta$, $\alpha + \beta + \gamma \le (k^2 + k + 1)\gamma$, which shows that

$$\alpha + \beta + \gamma \le (k^2 + k + 1)\min\{\alpha, \beta, \gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a + b + c)^{k^2+k+1}$. So abc divides $(a + b + c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a, b, c, the highest power of p that divides abc also divides $(a + b + c)^{k^2+k+1}$. It follows that abc divides $(a + b + c)^{k^2+k+1}$. ■

Solution 63. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α, β, γ such that $\alpha + \beta + \gamma = k^2 + k + 1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α, β, γ are all positive. It remains to consider the case when exactly one of α, β, γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \ge k + 1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \le k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \ge k + 1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \le k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \ge k + 1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \le k$$

holds. This shows that abc divides $(a + b + c)^{k^2+k+1}$. ■

Exercise 70 (RMO 2012c P2, , cf. RMO 2002 P3, RMO 2012d P2, RMO 2012a P2, RMO 2012b P2)**.** Let a, b, c be positive integers such that $a|b^5, b|c^5$ and $c|a^5$. Prove that abc divides $(a + b + c)^{31}$.

Solution 64. In the following, k denotes one of the integers $2, 3, 4, 5$. Assume that $a \mid b^k, b \mid c^k, c \mid a^k$. Note that if one of a, b, c is equal to 1, then all of them are equal to 1, and then there is nothing to prove. Henceforth, let us assume that a, b, c are larger than 1. The given conditions imply that the set of prime divisors of a, b, c are equal.

Let p be a common prime divisor of a, b, c. Let $p^\alpha, p^\beta, p^\gamma$ denote the highest powers of p dividing a, b, c respectively. The given divisibility conditions imply that

$$\alpha \le k\beta, \beta \le k\gamma, \gamma \le k\alpha.$$

This yields

$$\alpha + \beta + \gamma \le \alpha + k^2\alpha + 2\alpha = (k^2 + k + 1)\alpha,$$

and similarly, $\alpha + \beta + \gamma \le (k^2 + k + 1)\beta$, $\alpha + \beta + \gamma \le (k^2 + k + 1)\gamma$, which shows that

$$\alpha + \beta + \gamma \le (k^2 + k + 1)\min\{\alpha, \beta, \gamma\}.$$

Note that $p^{\min\{\alpha,\beta,\gamma\}}$ divides $a+b+c$. By the above inequality, $p^{\alpha+\beta+\gamma}$ divides $(a + b + c)^{k^2+k+1}$. So abc divides $(a + b + c)^{k^2+k+1}$. Also note that $p^{\alpha+\beta+\gamma}$ is the highest power of p dividing abc. This shows that for each common prime divisor p of a, b, c, the highest power of p that divides abc also divides $(a + b + c)^{k^2+k+1}$. It follows that abc divides $(a + b + c)^{k^2+k+1}$. ■

Solution 65. It suffices to show that abc divides $a^\alpha b^\beta c^\gamma$ for any nonnegative integers α, β, γ such that $\alpha + \beta + \gamma = k^2 + k + 1$. Note that

$$abc \mid ac^{k+1} \mid a^{k^2+k+1}, \quad abc \mid a^{k+1}b \mid b^{k^2+k+1}, \quad abc \mid b^{k+1}c \mid c^{k^2+k+1},$$

which shows that abc divides $a^{k^2+k+1}, b^{k^2+k+1}, c^{k^2+k+1}$. Moreover, abc divides $a^\alpha b^\beta c^\gamma$ when α, β, γ are all positive. It remains to consider the case when exactly one of α, β, γ is zero, which we assume from now on. If $\alpha = 0$, then

$$abc \mid b^{k+1}c \mid b^\beta c^\gamma \text{ if } \beta \ge k + 1,$$
$$abc \mid c^{k^2+1}b \mid b^\beta c^\gamma \text{ if } \beta \le k$$

holds. If $\beta = 0$, then

$$abc \mid c^{k+1}a \mid c^\gamma a^\alpha \text{ if } \gamma \ge k + 1,$$
$$abc \mid a^{k^2+1}c \mid c^\gamma a^\alpha \text{ if } \gamma \le k$$

holds. If $\gamma = 0$, then

$$abc \mid a^{k+1}b \mid a^\alpha b^\beta \text{ if } \alpha \ge k + 1,$$
$$abc \mid b^{k^2+1}a \mid a^\alpha b^\beta \text{ if } \alpha \le k$$

holds. This shows that abc divides $(a + b + c)^{k^2+k+1}$. ■

The content posted here and at this blog by Evan Chen are quite useful.

Exercise 71 (RMO 2003 P2,). If n is an integer greater than 7, prove that $\binom{n}{7} - \left[\frac{n}{7}\right]$ is divisible by 7. [Here $\binom{n}{7}$ denotes the number of ways of choosing 7 objects from among n objects; also for any real number x, $[x]$ denotes the greatest integer not exceeding x.]

Solution 66. If $x_1, x_2, \ldots, x_k$ are real numbers with $k \geq 2$ and $1 \leq \ell \neq k$ is an integer, then we denote by $x_1 \ldots \widehat{x_\ell} \ldots x_k$ the product

$$x_1 \ldots x_{\ell-1} x_{\ell+1} \ldots x_k.$$

Let n be an integer greater than 7. Let q, r be integers with $0 \leq r \leq 6$ and $q \geq 1$ such that n is equal to $7q+r$. We need to show that $\binom{7q+r}{7} - q$ is divisible by 7, which is equivalent to

$$6!\left(\binom{7q+r}{7} - q\right) \equiv 0 \bmod 7 \tag{4}$$

since 6! is coprime to 7. Note that $6!\binom{7q+r}{7}$ is equal to

$$\frac{(7q+r)(7q+r-1)(7q+r-2)(7q+r-3)(7q+r-4)(7q+r-5)(7q+r-6)}{7}.$$

It follows that

$$6!\binom{7q+r}{7} \equiv q \cdot r(r-1)\cdots\widehat{(r-r)}\cdots(r-6) \bmod 7 \equiv q \cdot 6! \bmod 7.$$

This shows that Eq. (4) holds, which proves the result. ■

Exercise 72 (RMO 2005 P2,). If x, y are integers and 17 divides both the expressions $x^2 - 2xy + y^2 - 5x + 7y$ and $x^2 - 3xy + 2y^2 + x - y$, then prove that 17 divides $xy - 12x + 15y$.

Walkthrough —

(a) Factorize $x^2 - 3xy + 2y^2 + x - y$ to show that

$$x \equiv y \pmod{17}, \quad \text{or } x \equiv 2y - 1 \pmod{17}$$

holds.

(b) Consider the above cases seperately, and use the divisibility of the other expression by 17 to obtain some congruence conditions on y. Using these conditions to read $xy - 12x + 15y$ modulo 17.

Solution 67. Let x, y be integers such that 17 divides both the expressions $x^2 - 2xy + y^2 - 5x + 7y$ and $x^2 - 3xy + 2y^2 + x - y$. Note that

$$x^2 - 3xy + 2y^2 + x - y = (x - y)(x - 2y + 1),$$

which is divisible by 17. It follows that

$$x \equiv y \bmod 17, \quad \text{or } x \equiv 2y - 1 \bmod 17$$

holds.

Let us consider the case that $x \equiv y \bmod 17$. It follows that

$$x^2 - 2xy + y^2 - 5x + 7y \equiv (x-y)^2 - 5x + 7y \equiv 2y \bmod 17.$$

Since 17 divides $x^2 - 2xy + y^2 - 5x + 7y$, we get $2y \equiv 0 \bmod 17$, which yields $x \equiv y \equiv 0 \bmod 17$, and hence 17 divides $xy - 12x + 15y$.

Let us consider the case that $x \equiv 2y - 1 \bmod 17$. Using $x^2 - 2xy + y^2 - 5x + 7y \equiv 0 \bmod 17$, we obtain

$$(2y-1)^2 - 2(2y-1)y + y^2 - 5(2y-1) + 7y \equiv 0 \bmod 17,$$

which yields $y^2 - 5y + 6 \equiv 0 \bmod 7$. This implies that $(y-2)(y-3) \equiv 0 \bmod 17$. This shows that either $x \equiv 3 \bmod 17, y \equiv 2 \bmod 17$ holds, or $x \equiv 5 \bmod 17, y \equiv 3 \bmod 17$ holds. If $x \equiv 3 \bmod 17, y \equiv 2 \bmod 17$ holds, then

$$xy - 12x + 15y \equiv 6 - 36 + 30 \equiv 0 \bmod 17$$

holds. If $x \equiv 5 \bmod 17, y \equiv 3 \bmod 17$ holds, then we obtain

$$xy - 12x + 15y \equiv 15 - 60 + 45 \equiv 0 \bmod 17.$$

This proves that 17 divides $xy - 12x + 15y$. ■

Exercise 73 (RMO 2006 P2,). Find the least possible value of $a + b$, where a, b are positive integers such that 11 divides $a + 13b$ and 13 divides $a + 11b$.

Solution 68. Let a, b be positive integers such that 11 divides $a + 13b$ and 13 divides $a + 11b$. It follows that 11 divides $6a + 78b = 6a + b + 77b$, and 13 divides $6a + 66b = 6a + b + 65b$. This shows that $6a + b$ is divisible by 11 and 13. Since the integers $11, 13$ are relatively prime, we obtain that $11 \cdot 13 = 143$ divides $6a + b$.

If $b \geq 4$, then we obtain

$$6a + 6b = 6a + b + 5b \geq 143 + 20 = 163,$$

which implies that

$$a + b \geq 28.$$

If $b = 1$, then $6a + b = 143$ admits no solutions in the integers, and hence $6a + b \geq 3 \cdot 143$ holds, which implies that $a + b \geq (6a + b)/6 > 28$. If $b = 2$, then $6a + b = 143$ admits no solutions in the integers, and hence $6a + b \geq 2 \cdot 143$ holds, which shows that $a + b \geq (6a + b)/6 > 28$. If $b = 3$, then $6a + b = 143$

admits no solutions in the integers, and hence $6a + b \geq 3 \cdot 143$ holds, which implies that $a + b \geq (6a + b)/6 > 28$. Considering these cases, it follows that $a + b \geq 28$.

Note that $a = 23, b = 5$ satisfy the given conditions. This shows that least possible value of $a + b$ accross all positive integers a, b satisfying the given divisibility conditions is 28. ■

Solution 69. Let a, b be positive integers such that 11 divides $a + 13b$, 13 divides $a + 11b$. It follows that $a + 2b$ is equal to $11m$, and $a - 2b$ is equal to $13n$ for some integers m, n with $m \geq 1$. This gives

$$a = \frac{11m + 13n}{2}, b = \frac{11m - 13n}{4}, a + b = \frac{33m + 13n}{4}.$$

Since b is an integer, we obtain $3m \equiv n \bmod 4$. If (m, n) is equal to $(3, 1)$, then note that

$$A := \frac{11m + 13n}{2}, B := \frac{11m - 13n}{4}$$

are positive integers satisfying

$$A + B = 23 + 5 = 28,$$

and 11 divides $A + 2B$ and 13 divides $A - 2B$, or equivalently, 11 divides $A + 13B$ and 13 divides $A + 11B$. This shows that $a + b \leq 28$. Since m is positive, we get

$$\frac{33 + 13n}{4} \leq \frac{33m + 13n}{4} = a + b \leq 28,$$
$$\frac{11 - 13n}{4} \leq \frac{11m - 13n}{4} = b < a + b \leq 28,$$

which gives $-7 \leq n \leq 6$. Note that if n is nonnegative, then we obtain

$$\frac{33m}{4} \leq a + b \leq 28,$$

which gives $m \leq 3$, and when n is negative, we have

$$\frac{33m + 13n}{4} \leq 28,$$

which gives

$$33m \leq 112 - 13n \leq 112 + 13 \cdot 7 = 203,$$

implying $m \leq 6$. Moreover, we have

$$m \geq \frac{13}{11}|n| > |n|.$$

Noting that $m + n \equiv 0 \bmod 4$ holds, it follows that the pair (m, n) is equal to one of $(3, 1), (6, -2), (5, -1)$. If (m, n) is equal to $(6, -2), (5, -1)$, then $b = \frac{11m - 13n}{4}$ is larger than 28. Consequently, (m, n) is equal to $(3, 1)$. This shows that the least value of $a + b$ is 28. ■

Example 74 (India BMath 2007 P1)**.** Let n be a positive integer. If n has odd number of divisors (other than 1 and n), then show that n is a perfect square.

Exercise 75 (RMO 2009 P3,)**.** Show that $3^{2008}+4^{2009}$ can be written as product of two positive integers each of which is larger than 2009^{182}.

Solution 70. Note that $3^{2008}+4^{2009}$ is equal to $4x^4+y^4$ where $x=4^{502}, y=3^{502}$. Also note that

$$3^{2008}+4^{2009}=(2x^2+y^2-2xy)(2x^2+y^2+2xy)$$

holds. Since $x\neq y$, it follows that the integers $2x^2+y^2+2xy, 2x^2+y^2-2xy$ are strictly larger than x^2. So it suffices to show $x^2\geq 2009^{182}$, which is equivalent to $x\geq 2009^{91}$, which follows since

$$x=4^{502}=2^{1004}>2^{11\cdot 91}=2048^{91}>2009^{91}$$

holds. ■

Exercise 76 (RMO 2011a P3,)**.** A natural number n is chosen strictly between two consecutive perfect squares. The smaller of these two squares is obtained by subtracting k from n and the larger by adding ℓ to n. Prove that $n-k\ell$ is a perfect square.

Solution 71. Let m be a positive integer such that

$$m^2=n-k, (m+1)^2=n+\ell$$

holds. This yields

$$\begin{aligned}n-k\ell&=n-(n-m^2)((m+1)^2-n)\\&=n+n^2-n(m^2+(m+1)^2)+m^2(m+1)^2\\&=n^2-n(2m^2+2m)+(m^2+m)^2\\&=(n-m^2-m)^2,\end{aligned}$$

which is a perfect square. ■

Example 77 (India Pre-RMO 2012)**.** Let $P(n)=(n+1)(n+3)(n+5)(n+7)$. What is the largest integer that is a divisor of $P(n)$ for all positive even integers n?

Solution 72. Note that

$$P(2)=3\cdot 5\cdot 7\cdot 9,$$

$$P(10) = 11 \cdot 13 \cdot 15 \cdot 17,$$
$$P(16) = 17 \cdot 19 \cdot 21 \cdot 23$$

holds. Also note that for any integer n, at least one of $n+1, n+3, n+5$ is divisible by 3. It follows that the largest integer that divides $P(n)$ for all positive even integers n is equal to 3. ■

Exercise 78 (RMO 2012f P2,). Let n be a positive integer such that 13 divides $n^2 + 3n + 51$. Show that 169 divides $21n^2 + 89n + 44$.

Solution 73. Note that the polynomial $X^2+3X+51$ is congruent to X^2+3X-1 modulo 13, which vanishes at 5 mod 13, and its discriminant is congruent to 0 modulo 13. This shows that the polynomial $X^2 + 3X + 51$ is congruent to $(X-5)^2$ modulo 13. So n is congruent to 5 mod 13.

Note that the discriminant of $21X^2 + 89X + 44$ satisfies

$$89^2 - 4 \cdot 21 \cdot 44 \equiv (-2)^2 - 4 \cdot (-5) \cdot 5 \equiv 4 \cdot 26 \bmod 13,$$

and hence this polynomial is congruent to $21(X-k)^2$ modulo 13 for some integer k. Observing that

$$21 \cdot 5^2 + 89 \cdot 5 + 44 \equiv (-5) \cdot (-1) - 2 \cdot 5 + 5 \equiv 0 \bmod 13,$$

it follows that k is congruent to 5 mod 13. This implies that $21X^2 + 89X + 44$ is congruent to $21(X-5)^2$ modulo 13. Since n is congruent to 5 mod 13, it follows that 169 divides $21n^2 + 89n + 44$. ■

Exercise 79 (RMO 2012e P6,). A computer program generated 175 positive integers at random, none of which had a prime divisor greater than 10. Prove that there are three numbers among them whose product is the cube of an integer.

Solution 74. Denote these integers by

$$2^{a_1}3^{b_1}5^{c_1}7^{d_1}, 2^{a_2}3^{b_2}5^{c_2}7^{d_2}, \cdots, 2^{a_{175}}3^{b_{175}}5^{c_{175}}7^{d_{175}}.$$

By pigeonhole principle, there exists a subset A of $\{1, 2, \ldots, 175\}$ of size 59 such that the integers a_i, for i in A, are congruent to each other modulo 3. By the same principle, it follows that there exists a subset B of A of size 20 such that the integers b_i, for i in B, are congruent to each other modulo 3. Applying the same principle, we obtain a subset of C of B of size 7 such that the integers c_i, for i in C, are congruent to each other modulo 3. Using the principle principle once again, we get a subset D of C of size 3 such that the integers d_i, for i in D, are congruent to each other modulo 3. Since $D \subseteq C \subseteq B \subseteq A$ holds, it follows that the product of the three integers

$$2^{a_i}3^{b_i}5^{c_i}7^{d_i}, i \in D$$

is a perfect cube. ■

Exercise 80 (RMO 2013d P2,). Determine the smallest prime that does not divide any five-digit number whose digits are in a strictly increasing order.

Solution 75. Note that 2 divides 12346, 3 and 5 divide 12345, and 7 divides 12348. Hence, the smallest prime satisfying the required condition is at least as large as 11.

Let $(abcde)_{10}$ be a five-digit number in base 10 with $1 \leq a < b < c < d < e \leq 9$. Note that

$$a - b + c - d + e = (a - b) + (c - d) + e \leq e \leq 9$$

holds, and

$$a - b + c - d + e = a + (c - b) + (e - d) \geq a \geq 1$$

holds. This shows that 11 does not divide $a - b + c - d + e$, or equivalently 11 does not divide the integer $(abcde)_{10}$. Since 11 does not divide any five-digit number whose digits are in a strictly increasing order, it follows that the smallest prime satisfying the required condition is at most 11.

We conclude that 11 is the smallest prime satisfying the required condition. ■

Exercise 81 (RMO 2013e P5,). Let a_1, b_1, c_1 be natural numbers. We define

$$a_2 = \gcd(b_1, c_1), \quad b_2 = \gcd(c_1, a_1), \quad c_2 = \gcd(a_1, b_1)$$

and

$$a_3 = \operatorname{lcm}(b_2, c_2), \quad b_3 = \operatorname{lcm}(c_2, a_2), \quad c_3 = \operatorname{lcm}(a_2, b_2).$$

Show that $\gcd(b_3, c_3) = a_2$.

Solution 76. It suffices to show that the highest power of p dividing $\gcd(b_3, c_3)$ and a_2 are equal for any prime p. Fix a prime number p. Let p^x, p^y, p^z denote the highest powers of p dividing a_1, b_1, c_1 respectively. So the highest exponents of p dividing a_2, b_2, c_2 are

$$\min\{y, z\}, \min\{z, x\}, \min\{x, y\}$$

respectively. Thus the highest exponents of p dividing b_3, c_3 are

$$\max\{\min\{x, y\}, \min\{y, z\}\}, \max\{\min\{y, z\}, \min\{z, x\}\}$$

respectively. So it remains to show that

$$\min\{\max\{\min\{x, y\}, \min\{y, z\}\}, \max\{\min\{y, z\}, \min\{z, x\}\}\} = \min\{y, z\} \tag{5}$$

holds. Note that $\min\{x, y\}, \min\{y, z\}$ are less than or equal to y, which gives

$$\max\{\min\{x, y\}, \min\{y, z\}\} \leq y.$$

Similarly, it follows that

$$\max\{\min\{y, z\}, \min\{z, x\}\} \leq z.$$

This gives that

$$\min\{\max\{\min\{x, y\}, \min\{y, z\}\}, \max\{\min\{y, z\}, \min\{z, x\}\}\} \leq \min\{y, z\}.$$

Note also that

$$\begin{aligned}\max\{\min\{x, y\}, \min\{y, z\}\} &\geq \min\{y, z\},\\ \max\{\min\{y, z\}, \min\{z, x\}\} &\geq \min\{y, z\}\end{aligned}$$

holds, which yields

$$\min\{\max\{\min\{x, y\}, \min\{y, z\}\}, \max\{\min\{y, z\}, \min\{z, x\}\}\} \geq \min\{y, z\}.$$

This proves that Eq. (5) holds. ■

Exercise 82 (RMO 2014c P3,). Prove that for any natural number $n < 2310$, $n(2310 - n)$ is not divisible by 2310.

Solution 77. Suppose $n < 2310$ is a positive integer such that $n(2310 - n)$ is equal to $2310k$ for some positive integer k. It follows that there are positive integers a, b such that a divides n, b divides $2310 - n$ and $ab = 2310$. Note that b also divides n. This shows that n is divisible by the least common multiple of the integers a and b, that is, by $ab/\gcd(a, b)$. Note that

$$2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$$

holds, and hence 2310 is a square-free integer. Since ab is equal to 2310, it follows that $\gcd(a, b) = 1$. This shows that ab divides n, that is, 2310 divides n, which is impossible. This completes the proof. ■

Exercise 83 (RMO 2014b P3,). Find all pairs of (x, y) of positive integers such that $2x + 7y$ divides $7x + 2y$.

Solution 78. Let (x, y) be a pair of positive integers such that $2x + 7y$ divides $7x+2y$. Let d denote the positive integer such that $7x+2y$ is equal to $(2x+7y)d$. This gives

$$7(x - dy) = 2(dx - y).$$

Since the integers $2, 7$ are relatively prime, it follows that 2 divides $x - dy$ and 7 divides $dx - y$. This shows that

$$\frac{x - dy}{2} = \frac{dx - y}{2} = k$$

holds for some integer k. This shows that

$$x - dy = 2k, \quad dx - y = 7k,$$

which gives

$$(d^2 - 1)x = (7d - 2)k, \quad (d^2 - 1)y = (7 - 2d)k.$$

Since y is positive, we obtain $d \leq 3$. Note that if $d = 1$, then $k = 0$ and hence $x = y$. If $d = 2$, then (x, y) is equal to $(4k, k)$, and if $d = 3$, then (x, y) is equal to $(\frac{19}{8}k, \frac{1}{8}k)$, in which case k is a multiple of 8. So (x, y) is an element of the set

$$\bigcup_{r=1}^{\infty} \{(r, r), (4r, r), (19r, r)\}.$$

Also note that for any element (a, b) of this set, the integer $2a + 7b$ divides $7a + 2b$. This proves that the required pairs form the above set. ■

Exercise 84 (RMO 2014b P4,). For any positive integer $n > 1$, let $P(n)$ denote the largest prime not exceeding n. Let $N(n)$ denote the next prime larger than $P(n)$. (For example $P(10) = 7$ and $N(10) = 11$, while $P(11) = 11$ and $N(11) = 13$.) If $n + 1$ is a prime number, prove that the value of the sum

$$\frac{1}{P(2)N(2)} + \frac{1}{P(3)N(3)} + \frac{1}{P(4)N(4)} + \cdots + \frac{1}{P(n)N(n)} = \frac{n - 1}{2n + 2}.$$

Solution 79. Let p_n denote the n-th prime for $n \geq 1$. Note that for any integer m with $p_n \leq m \leq p_{n+1} - 1$, the integer $P(m)N(m)$ is equal to $p_n p_{n+1}$. This gives

$$\sum_{m=p_n}^{p_{n+1}-1} \frac{1}{P(m)N(m)} = \frac{p_{n+1} - p_n}{p_n p_{n+1}} = \frac{1}{p_n} - \frac{1}{p_{n+1}}.$$

Let k denote the positive integer such that $n + 1$ is equal to p_k. Note that

$$\begin{aligned}
&\frac{1}{P(2)N(2)} + \frac{1}{P(3)N(3)} + \frac{1}{P(4)N(4)} + \cdots + \frac{1}{P(n)N(n)} \\
&= \sum_{m=p_1}^{p_2-1} \frac{1}{P(m)N(m)} + \sum_{m=p_2}^{p_3-1} \frac{1}{P(m)N(m)} + \cdots + \sum_{m=p_{k-1}}^{p_k-1} \frac{1}{P(m)N(m)} \\
&= \left(\frac{1}{p_1} - \frac{1}{p_2}\right) + \left(\frac{1}{p_2} - \frac{1}{p_3}\right) + \cdots + \left(\frac{1}{p_{k-1}} - \frac{1}{p_k}\right)
\end{aligned}$$

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$$= \frac{1}{p_1} - \frac{1}{p_k}$$
$$= \frac{1}{2} - \frac{1}{n+1}$$
$$= \frac{n-1}{2n+2}.$$

■

Exercise 85 (RMO 2014d P3,). Determine all pairs $m > n$ of positive integers such that

$$1 = \gcd(n+1, m+1) = \gcd(n+2, m+2) = \cdots = \gcd(m, 2m-n).$$

Solution 80. Let $m > n$ be positive integers satisfying the given condition. It follows that

$$1 = \gcd(n+1, m-n) = \gcd(n+2, m-n) = \cdots = \gcd(m, m-n).$$

Note that the positive integer $m-n$ is relatively prime to the $m-n$ consecutive integers

$$n+1, n+2, \ldots, m.$$

This implies that $m - n = 1$.

Note that if $m - n = 1$ holds for two positive integers, then

$$1 = \gcd(n+1, m-n) = \gcd(n+2, m-n) = \cdots = \gcd(m, m-n)$$

holds, and which yields

$$1 = \gcd(n+1, m+1) = \gcd(n+2, m+2) = \cdots = \gcd(m, 2m-n).$$

This proves that the pairs (m, n) satisfying the given condition are precisely of the pairs of the form $(n+1, n)$ as n ranges over the positive integers. ■

Exercise 86 (RMO 2016d P3,). The present ages in years of two brothers A and B, and their father C are three distinct positive integers a, b, and c respectively. Suppose $\frac{b-1}{a-1}$ and $\frac{b+1}{a+1}$ are two consecutive integers, and $\frac{c-1}{b-1}$ and $\frac{c+1}{b+1}$ are two consecutive integers. If $a + b + c \leq 150$ determine a, b and c.

Solution 81. Note that

$$\frac{b-1}{a-1} - \frac{b+1}{a+1} = \frac{(a+1)(b-1) - (a-1)(b+1)}{a^2-1} = \frac{2(b-a)}{a^2-1}$$

holds, and

$$\frac{c-1}{b-1} - \frac{c+1}{b+1} = \frac{(b+1)(c-1) - (b-1)(c+1)}{b^2-1} = \frac{2(c-b)}{b^2-1}$$

holds. Note that if $a > b$, then

$$\frac{2(a-b)}{a^2-1} = 1$$

holds, which yields

$$2 - 2b = (a-1)^2,$$

which gives $a = 2$, and $b = 0$, which is impossible since $b \geq 1$. This shows that $a \leq b$. It follows that

$$\frac{2(b-a)}{a^2-1} = 1, \quad \frac{2(c-b)}{b^2-1} = 1$$

hold. This yields

$$2(b-a) = a^2 - 1,$$
$$2(c-b) = b^2 - 1,$$

which implies

$$2b = a^2 + 2a - 1,$$
$$2c = b^2 + 2b - 1.$$

Note that a, b are odd, and

$$2(b+1) = (a+1)^2,$$
$$2(c+1) = (b+1)^2$$

holds. This shows that

$$a + b + c + 3 = a + 1 + \frac{(a+1)^2}{2} + \frac{(a+1)^4}{8}.$$

Using the bound $a + b + c \leq 150$, we obtain

$$a + 1 + \frac{(a+1)^2}{2} + \frac{(a+1)^4}{8} \leq 153,$$

which shows that $a \leq 4$. Since a is odd, it follows that $a = 3$. It follows that

$$b = 7, \quad c = 31.$$

Note that the triple $(a, b, c) = (3, 7, 31)$ satisfies $a + b + c \leq 150$, and $\frac{b-1}{a-1} = 3$ and $\frac{b+1}{a+1} = 2$ are two consecutive integers, and $\frac{c-1}{b-1} = 5$ and $\frac{c+1}{b+1} = 4$ are two consecutive integers.

This proves that

$$a = 3, \quad b = 7, \quad c = 31.$$

■

Exercise 87 (RMO 2016c P3,). Let a, b, c, d, e, f be positive integers such that

$$\frac{a}{b} < \frac{c}{d} < \frac{e}{f}.$$

Suppose $af - be = -1$. Show that $d \geq b + f$.

Solution 82. The given conditions yield

$$ad < bc, \quad cf < de.$$

This gives

$$adf \leq bcf - f \leq bde - b - f,$$

which implies

$$d(be - af) \geq b + f.$$

It follows that $d \geq b + f$. ■

Exercise 88 (RMO 2016g P3,). a, b, c, d are integers such that $ad + bc$ divides each of a, b, c and d. Prove that $ad + bc = \pm 1$.

> **Remark.** What happens when a, b, c, d are equal to zero?

Solution 83. Note that there exist integers p, q, r, s such that

$$\begin{aligned} a &= p(ad + bc), \\ b &= q(ad + bc), \\ c &= r(ad + bc), \\ d &= s(ad + bc) \end{aligned}$$

holds. This yields

$$ad + bc = (ps + qr)(ad + bc)^2.$$

If $ad + bc = 0$, then all of the integers a, b, c, d are equal to zero, which is impossible. This shows that $ad + bc$ is nonzero, and hence, we obtain

$$(ps + qr)(ad + bc) = 1,$$

which implies

$$ad + bc = \pm 1.$$

■

Exercise 89 (RMO 2016g P6,). Positive integers a, b, c satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} < 1$. Prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{41}{42}$. Also prove that equality in fact holds in the second inequality.

Solution 84. Let a, b, c be positive integers satisfying

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c}<1. \tag{6}$$

Since $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ is symmetric with respect to a, b, c, by reordering them (if necessary), we may (and do) assume that $a \leq b \leq c$. Using Eq. (6), it follows that $a \geq 2$.

Let us consider the case that $a = 3$. Using Eq. (6) it follows that one of b, c is greater than 3, and hence

$$\begin{aligned}\frac{1}{3}+\frac{1}{3}+\frac{1}{4} &= \frac{2}{3}+\frac{1}{4}\\ &= \frac{9}{12}\\ &= \frac{3}{4}\\ &< \frac{41}{42}\end{aligned}$$

holds.

Now, let us consider the case that $a = 2$. Using Eq. (6), it follows that the integers b, c are greater than or equal to 3. If $b = 3$, then using Eq. (6), it follows that $c > 6$, which shows that

$$\begin{aligned}\frac{1}{a}+\frac{1}{b}+\frac{1}{c} &= \frac{1}{2}+\frac{1}{3}+\frac{1}{c}\\ &\leq \frac{5}{6}+\frac{1}{7}\\ &= \frac{41}{42}\end{aligned}$$

holds. If $b = 4$, then using Eq. (6), it follows that $c > 4$, which shows that

$$\begin{aligned}\frac{1}{a}+\frac{1}{b}+\frac{1}{c} &= \frac{1}{2}+\frac{1}{4}+\frac{1}{c}\\ &\leq \frac{3}{4}+\frac{1}{5}\\ &= \frac{19}{20}\\ &\leq \frac{41}{42}\end{aligned}$$

holds. If $b \geq 5$ holds, then using Eq. (6), it follows that

$$\begin{aligned}\frac{1}{a}+\frac{1}{b}+\frac{1}{c} &\leq \frac{1}{2}+\frac{2}{b}\\ &\leq \frac{1}{2}+\frac{2}{5}\end{aligned}$$

The content posted here and at this blog by Evan Chen are quite useful.

$$= \frac{9}{10}$$
$$\leq \frac{41}{42}$$

holds. This proves the inequality

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{41}{42}.$$

Observing that

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{7} = \frac{41}{42}$$

holds, it follows that equality holds in the above inequality for some choice of a, b, c. ■

Exercise 90 (RMO 2016e P2,). Consider a sequence $(a_k)_{k\geq 1}$ of natural numbers defined as follows: $a_1 = a$ and $a_2 = b$ with $a, b > 1$ and $\gcd(a, b) = 1$ and for all $k > 0$, $a_{k+2} = a_{k+1} + a_k$. Prove that for all natural numbers n and k, $\gcd(a_n, a_{n+k}) < \frac{a_k}{2}$.

Remark. Note that for the above inequality, it is necessary that $a_1 \geq 3$. In fact, **if $a_1 = 2$, then the above inequality is not strict for $k = 1$.**

Solution 85. Consider the sequence $\{f_n\}_{n\geq 1}$ satisfying the recurrence relation

$$f_{n+2} = f_{n+1} + f_n, \quad \text{for any } n \geq 1,$$

and $f_1 = 0, f_2 = 1$. Note that

$$a_n = af_{n-1} + bf_n, \quad \text{for any } n \geq 2.$$

Claim — For any integers $n \geq 2, k \geq 1$,

$$f_{n+k-1} = f_{n-1}f_k + f_nf_{k+1}$$

holds.

Proof of the Claim. It suffices to consider the case $k \geq 2$. Note that

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} f_n & f_{n+1} \\ f_{n+1} & f_{n+2} \end{pmatrix}$$

holds for $n = 1$, and if it holds for $n = m$ for some integer $m \geq 1$, then it holds for $n = m + 1$ since

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^{m+1} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} f_m & f_{m+1} \\ f_{m+1} & f_{m+2} \end{pmatrix}$$

$$= \begin{pmatrix} f_{m+1} & f_{m+2} \\ f_{m+1}+f_m & f_{m+2}+f_{m+1} \end{pmatrix}$$
$$= \begin{pmatrix} f_{m+1} & f_{m+2} \\ f_{m+2} & f_{m+3} \end{pmatrix}$$

holds. This shows that the above holds for any positive integer n. Consequently, for any integers $n, k \geq 2$, it follows that

$$\begin{pmatrix} f_{n-1} & f_n \\ f_n & f_{n+1} \end{pmatrix} \begin{pmatrix} f_k & f_{k+1} \\ f_{k+1} & f_{k+2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^{n-1+k} = \begin{pmatrix} f_{n+k-1} & f_{n+k} \\ f_{n+k} & f_{n+k+1} \end{pmatrix},$$

which yields

$$f_{n+k-1} = f_{n-1}f_k + f_n f_{k+1}.$$

□

Let n, k be positive integers. Using the above Claim, we obtain

$$\begin{aligned} a_{n+k} &= af_{n+k-1} + bf_{n+k} \\ &= a(f_{n-1}f_k + f_n f_{k+1}) + b(f_{n-1}f_{k+1} + f_n f_{k+2}) \\ &= a(f_{n-1}f_k + f_n f_{k+1}) + b(f_{n-1}f_{k+1} + f_n f_k + f_n f_{k+1}) \\ &= (af_{n-1} + bf_n)f_k + (af_n + bf_{n-1} + bf_n)f_{k+1} \\ &= a_n f_k + (af_n + bf_{n+1})f_{k+1} \\ &= a_n f_k + a_{n+1}f_{k+1}. \end{aligned}$$

This yields

$$\begin{aligned} \gcd(a_n, a_{n+k}) &= \gcd(a_n, a_n f_k + a_{n+1}f_{k+1}) \\ &= \gcd(a_n, a_{n+1}f_{k+1}). \end{aligned}$$

Note that

$$\gcd(a_n, a_{n+1}) = 1$$

holds for $n = 1$, and if it holds for $n = m$ for some positive integer m, then

$$\begin{aligned} \gcd(a_{m+1}, a_{m+2}) &= \gcd(a_{m+1}, a_{m+1} + a_m) \\ &= \gcd(a_{m+1}, a_m) \\ &= 1, \end{aligned}$$

which shows that the integers a_n, a_{n+1} are relatively prime for any positive integer n. If $a_1 \geq 3$, then it follows that

$$\gcd(a_n, a_{n+1}) = 1 < \frac{a_1}{2},$$

that is, for $k = 1$, the inequality

$$\gcd(a_n, a_{n+k}) < \frac{a_k}{2}$$

holds. Moreover, if $k \geq 2$, then using the above, we obtain

$$\begin{aligned} \gcd(a_n, a_{n+k}) &= \gcd(a_n, f_{k+1}) \\ &\leq f_{k+1} \\ &= f_{k-1} + f_k \\ &< \frac{af_{k-1} + bf_k}{2} \\ &= \frac{a_k}{2}. \end{aligned}$$

This completes the proof. ■

Exercise 91 (RMO 2017a P2,). Show that the equation

$$a^3 + (a+1)^3 + \cdots + (a+6)^3 = b^4 + (b+1)^4$$

has no solutions in integers a, b.

> **Walkthrough** — Read it modulo a suitable integer ★. Are there any intuitions about the choice of ★?

Solution 86. Note that the fourth powers of the integers $0, 1, 2, 3, 4, 5, 6$ are congruent to $0, 1, 2, 4, 4, 2, 1$ modulo 7 respectively. The shows that the sum of the fourth powers of two consecutive integers is congruent to one of

$$0+1, 1+2, 2+4, 4+2, 2+1, 1+0$$

modulo 7. Hence, the sum of the fourth powers of two consecutive integers is not divisible by 7. Also note the cubes of seven consecutive integers is congruent to

$$0^3 + 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3$$

modulo 7, which is congruent to

$$1^3 + 2^3 + 3^3 + (-3)^3 + (-2) + (-1)^3 = 0$$

modulo 7. This shows that the sum of the cubes of seven consecutive integers is not equal to the sum of the fourth powers of two consecutive integers. This completes the proof. ■

Exercise 92 (Mongolian Mathematical Olympiad 2025 Grades 9-10 E2,). Show that the equation $x^4 + x^3 + x^2 + x = y^6 + 61$ has no integer solutions.

Walkthrough —

(a) Read it modulo 7.

(b) For any integer y, show that $y^6 + 61$ is congruent to one of $5, 6$ modulo 7.

(c) For any integer x, determine the possible values of $x^4 + x^3 + x^2 + x$ modulo 7.

Exercise 93 (RMO 2017b P2)**.** For any positive integer n, let $d(n)$ denote the number of positive divisors of n; and let $\varphi(n)$ denote the number of elements from the set $\{1, 2, \ldots, n\}$ that are coprime to n. (For example, $d(12) = 6$ and $\varphi(12) = 4$.) Find the smallest positive integer n such that $d(\varphi(n)) = 2017$.

Solution 87. Let n be the least among the positive integers (if any) satisfying $d(\varphi(n)) = 2017$. Observe that $\varphi(n)$ is not equal to 1. Let $\varphi(n) = p_1^{\alpha_1} \ldots p_r^{\alpha_r}$ denote the factorization of $\varphi(n)$ into the product of the powers of distinct primes. We obtain

$$(\alpha_1 + 1) \ldots (\alpha_r + 1) = 2017.$$

Noting that 2017 is a prime, it follows that $r = 1$, and $\alpha_1 = 2016$, that is, $\varphi(n)$ is the 2016-th power of a prime. Since $\varphi(n)$ is larger than 1, it follows that it is an even number, and hence, $\varphi(n)$ is equal to 2^{2016}. Note that if an odd prime q divides n, then q^2 does not divide n, and $q - 1$ is a power of 2. It follows that n is equal to $2^k p_1 \ldots p_m$ where k, m are nonnegative integers, and $p_1 < \cdots < p_m$ are distinct odd primes, and $p_i - 1$ is a power of 2 for any $1 \le i \le m$.

Claim — If p is an odd prime such that $p - 1$ is a power of 2, then p is equal to $2^{2^i} + 1$ for some integer $i \ge 0$.

Proof of the Claim. It suffices to consider the case $p \ge 5$. Write $p = 2^a + 1$. Note that $a \ge 2$. If a is odd, then p is divisible by 3, and $p > 3$, which is a contradiction to the primality of p. This shows that a is even. If a is not a power of 2, then writing $a = 2^b c$ for some positive integers b, c, with c an odd integer, it follows that p is divisible by $2^{2^b} + 1$, which shows that c is equal to 1. This completes the proof. □

Note that the above Claim shows that for any $1 \le i \le m$, p_i is equal to $2^{2^{a_i}} + 1$ for some nonnegative integer a_i.

Claim — For no integer $1 \le i \le m$, the prime p_i is equal to $2^{2^5} + 1$.

Proof of the Claim. Note that $641 = 5 \cdot 2^7 + 1$ is a prime. This gives $5 \cdot 2^7 \equiv -1 \bmod 641$, which yields $5^4 \cdot 2^{28} \equiv 1 \bmod 641$. Using $641 = 2^4 + 5^4$, it follows that $2^{32} + 1$ is divisible by 641. Since $2^{32} + 1$ is larger than $5 \cdot 2^7 + 1 = 641$, the Claim follows. □

Let us first consider the case that $k = 0$. Since $\varphi(n) = 2^{2016} < 2^{2^{11}}$ holds, it follows that

$$p_m \leq 2^{2^{10}} + 1.$$

The first Claim shows that $\{p_1, \ldots, p_m\}$ is a subset of

$$\left\{2^{2^i} + 1 \,|\, 0 \leq i \leq 10\right\}.$$

This shows that

$$\begin{aligned} 2^{2016} &= \varphi(p_1 \ldots p_m) \\ &\leq \prod_{0 \leq i \leq 10, i \neq 5} 2^{2^i} \quad \text{(using the second Claim)} \\ &= 2^{1+2+4+8+16+64+128+256+512+1024} \\ &= 2^{2015}, \end{aligned}$$

which is impossible.

Let us consider the case that $k \geq 1$. If $m \geq 1$, then note that

$$n > 2^k(p_1 - 1)(p_2 - 1) \ldots (p_m - 1)$$

and

$$\varphi(n) = \varphi(2^k(p_1 - 1)(p_2 - 1) \ldots (p_m - 1))$$

hold, which contradicts the minimality of n. This shows that $m = 0$, that is, n is a power of 2, and hence, $n = 2^{2017}$.

Noting that

$$d(\varphi(2^{2017})) = d(2^{2016}) = 2017$$

holds, it follows that the smallest positive integer n satisfying $d(\varphi(n)) = 2017$ is 2^{2017}. ■

Exercise 94 (RMO 2018a P5,). Find all natural numbers n such that $1 + [\sqrt{2n}]$ divides $2n$. (For any real number x, $[x]$ denotes the largest integer not exceeding x.)

Solution 88. Let n be a positive integer such that $1 + [\sqrt{2n}]$ divides $2n$. Write

$$2n = m^2 + k$$

where m is a positive integer, and k is an integer satisfying $0 \leq k \leq 2m$. Note that

$$1 + [\sqrt{2n}] = m + 1$$

holds. Since $1 + [\sqrt{2n}]$ divides $2n$, it follows that

$$\frac{2n}{1 + [\sqrt{2n}]} = \frac{m^2 + k}{m + 1}$$

$$= \frac{(m+1)^2 - 2(m+1) + k + 1}{m+1}$$
$$= m - 1 + \frac{k+1}{m+1}$$

is an integer, which shows that $m+1$ divides $k+1$. Using $0 \le k \le 2m$, we obtain $k = m$, which yields

$$n = \frac{m(m+1)}{2}.$$

Conversely, if n is equal to

$$\frac{m(m+1)}{2}$$

for some positive integer m, then $1 + [\sqrt{2n}]$ divides $2n$.

This proves that the positive integers satisfying the given condition are precisely the integers of the form

$$\frac{m(m+1)}{2}$$

for some positive integer m. ■

Exercise 95 (RMO 2018a P3,). For a rational number r, its *period* is the length of the smallest repeating block in its decimal expansion, for example, the number $r = 0.123123123\ldots$ has period 3. If S denotes the set of all rational numbers of the form $r = 0.\overline{abcdefgh}$ having period 8, find the sum of all elements in S.

Solution 89. The sum of all rational numbers of the form $0.\overline{abcdefgh}$ is equal to

$$\sum_{0 \le i_1, i_2, \ldots, i_8 \le 9} \left(\frac{i_1}{10} + \frac{i_2}{10^2} + \cdots + \frac{i_8}{10^8} \right) \frac{1}{1 - \frac{1}{10^8}}.$$

Note that the rational numbers of the form $0.\overline{abcdefgh}$ which have period less than 8, are of period 1 or 2 or 4, and hence, they are of the form $0.\overline{abcd}$. The sum of such rational numbers is equal to

$$\sum_{0 \le i_1, i_2, i_3, i_4 \le 9} \left(\frac{i_1}{10} + \frac{i_2}{10^2} + \cdots + \frac{i_4}{10^4} \right) \frac{1}{1 - \frac{1}{10^4}}.$$

It follows that the sum of the rational numbers of the given form having period 8 is equal to

$$\sum_{0 \le i_1, i_2, \ldots, i_8 \le 9} \left(\frac{i_1}{10} + \frac{i_2}{10^2} + \cdots + \frac{i_8}{10^8} \right) \frac{1}{1 - \frac{1}{10^8}}$$

$$
\begin{aligned}
&- \sum_{0\le i_1,i_2,i_3,i_4\le 9} \left(\frac{i_1}{10}+\frac{i_2}{10^2}+\cdots+\frac{i_4}{10^4}\right)\frac{1}{1-\frac{1}{10^4}} \\
&= 10^7\left(\sum_{i=0}^{9} i\right)\left(\frac{1}{10}+\frac{1}{10^2}+\cdots+\frac{1}{10^8}\right)\frac{1}{1-\frac{1}{10^8}} \\
&- 10^3\left(\sum_{i=0}^{9} i\right)\left(\frac{1}{10}+\frac{1}{10^2}+\cdots+\frac{1}{10^4}\right)\frac{1}{1-\frac{1}{10^4}} \\
&= \frac{10^7}{9}\left(\sum_{i=0}^{9} i\right) - \frac{10^3}{9}\left(\sum_{i=0}^{9} i\right) \\
&= 5\times 10^7 - 5\times 10^3 \\
&= (50000-5)\times 10^3 \\
&= 49995000.
\end{aligned}
$$

■

Solution 90. Note that the rational numbers of the form $0.\overline{abcdefgh}$, not having period 8, have period 1 or 2 or 4, and hence, are of the form $0.\overline{abcd}$. This shows that there are precisely $10^8 - 10^4$ rational numbers of the form $0.\overline{abcdefgh}$, having period 8.

Observe that

$$0.\overline{abcdefgh} \mapsto 0.\overline{\mathsf{abcdefgh}},$$

where $\mathsf{x} = 9 - x$ for any $x \in \{a, b, c, d, e, f, g, h\}$, defines a map of order two from S to S, and this map has no fixed points. This shows that each element of S can be paired with its image under this map, and it produces $\frac{1}{2}|S|$ many pairs. Note that the sum of the elements of any such pair is equal to

$$0.\overline{999999} = 1.$$

Consequently, the sum of the elements of S is equal to

$$\frac{1}{2}|S| = \frac{10^8 - 10^4}{2} = 49995000.$$

■

Exercise 96 (RMO 2019b P1,). For each $n \in \mathbb{N}$ let d_n denote the greatest common divisor of n and $(2019 - n)$. Find value of $d_1 + d_2 + \cdots + d_{2018} + d_{2019}$.

Solution 91. Note that for a positive integer n, the greatest common divisor of n and $2019 - n$ is equal to the greatest common divisor of n and 2019. Also note that $2019 = 3 \cdot 673$, and that $3, 673$ are primes. This shows that for

an integer n satisfying $1 \leq n < 2019$, the greatest common divisor of n and $2019 - n$ is equal to

$$\begin{cases} 1 & \text{if } n \text{ is relatively prime to 3 and to 673,} \\ 3 & \text{if 3 divides } n, \\ 673 & \text{if 673 divides } n. \end{cases}$$

This yields

$$\begin{aligned} & d_1 + d_2 + \cdots + d_{2018} + d_{2019} \\ &= \sum_{\substack{1 \leq n < 2019, \\ \gcd(n, 2019) = 1}} 1 + \sum_{\substack{1 \leq n < 2019, \\ 3 \mid n}} 3 + \sum_{\substack{1 \leq n < 2019, \\ 673 \mid n}} 673 + 2019 \\ &= 2019 - \frac{2019}{3} - \frac{2019}{673} + \frac{2019}{3 \cdot 673} + 3 \cdot 672 + 673 \cdot 2 + 3 \cdot 673 \\ &= 2 \cdot 672 + 3 \cdot 672 + 5 \cdot 673 \\ &= 5 \cdot 1345 \\ &= 6725. \end{aligned}$$

■

Alternative solution. Note that for a positive integer n, the greatest common divisor of n and $2019 - n$ is equal to the greatest common divisor of n and 2019. Also note that $2019 = 3 \cdot 673$, and that $3, 673$ are primes. It follows that

$$\begin{aligned} & d_1 + d_2 + \cdots + d_{2019} \\ &= \gcd(1, 2019) + \gcd(2, 2019) + \cdots + \gcd(2019, 2019) \\ &= (\gcd(1, 3) + \gcd(2, 3) + \gcd(3, 3))\,(\gcd(1, 673) + \gcd(2, 673) + \cdots + \gcd(673, 673)) \\ &= (1 + 1 + 3)(1 + 1 + \cdots + 1 + 673) \\ &= 5 \cdot (672 + 673) \\ &= 5 \cdot 1345 \\ &= 6725. \end{aligned}$$

□

Exercise 97 (RMO 2023a P2,). Given a prime number p such that $2p$ is equal to the sum of the squares of some four consecutive positive integers. Prove that $p - 7$ is divisible by 36.

> **Walkthrough** — Show that the sum of four consecutive squares is congruent to 6 modulo 8, and conclude that $p \equiv 3 \bmod 4$. Considering congruence conditions modulo 3, prove that the smallest of the four consecutive numbers is a multiple

of 3. Deduce that the sum of the four consecutive squares is 5 modulo 9.

Solution 92. Let p be a prime satisfying the given condition. Write

$$2p = x^2 + (x+1)^2 + (x+2)^2 + (x+3)^2$$

where x is a positive integer. Note that

$$\begin{aligned} x^2 + (x+1)^2 + (x+2)^2 + (x+3)^2 &= 4x^2 + 12x + 14 \\ &\equiv 4x(x+1) + 6 \bmod 8 \\ &\equiv 6 \bmod 8 \end{aligned}$$

holds. It follows that $2p$ is congruent to 6 modulo 8, which shows that p is congruent to 3 modulo 4.

Note that if $x \equiv \pm 1 \bmod 3$, then

$$\begin{aligned} x^2 + (x+1)^2 + (x+2)^2 + (x+3)^2 &= 4x^2 + 12x + 14 \\ &\equiv x^2 + 2 \bmod 3 \\ &\equiv 0 \bmod 3 \end{aligned}$$

holds, which shows that 3 divides $2p$, and hence $p = 3$. However, 6 cannot be expressed as the sum of four consecutive positive integers since $3^2 > 6$. This shows that x is divisible by 3. It follows that

$$\begin{aligned} x^2 + (x+1)^2 + (x+2)^2 + (x+3)^2 &= 4x^2 + 12x + 14 \\ &\equiv 14 \bmod 9. \end{aligned}$$

So, $2p - 14$ is a multiple of 9, and hence, it is an even multiple of 9, implying that $p - 7$ is a multiple of 9. Since $p \equiv 3 \bmod 4$, we obtain $p \equiv 7 \bmod 36$. ■

Exercise 98 (RMO 2023b P1,). Let $\mathbb{N}$ be the set of all positive integers and

$$S = \left\{(a,b,c,d) \in \mathbb{N}^4 : a^2 + b^2 + c^2 = d^2\right\}.$$

Find the largest positive integer m such that m divides $abcd$ for all $(a,b,c,d) \in S$.

Walkthrough —

(a) Show that $(1,2,2,3)$ lies in S, and deduce that m divides 12.

(b) Let (a,b,c,d) be an element of S. Show that at least one of a,b,c,d is divisible by 3, and at least one of them is even.

(c) Prove that if d is even, then at least one of a,b,c is even, and that if d is odd, then at least two of a,b,c are even.

(d) Conclude that m is divisible by $2 \cdot 2 \cdot 3$.

Solution 93. Let m denote the largest positive integer such that it divides $abcd$ for all (a, b, c, d) in S. Note that $1^2 + 2^2 + 2^2 = 3^2$ holds, which shows that $(1, 2, 2, 3)$ lies in S. This shows that m divides 12.

Let (a, b, c, d) be an element of S. Note that at least one of a, b, c, d is divisible by 3 since

$$(\pm 1)^2 + (\pm 1)^2 + (\pm 1)^2 \not\equiv (\pm 1)^2 \bmod 3.$$

Also note that if all of a, b, c, d are odd, then we obtain

$$(\pm 1)^2 + (\pm 1)^2 + (\pm 1)^2 \equiv (\pm 1)^2 \bmod 4,$$

which is impossible. It follows that at least one of a, b, c, d is even.

If d is even, then at least one of a, b, c is even because the sum of squares of three odd integers is congruent to 3 modulo 4. If d is odd, then at least one of a, b, c is even. Note that $0 + (\pm 1)^2 + (\pm 1)^2 \not\equiv (\pm 1)^2 \bmod 4$, which shows that if d is odd, then at least two of a, b, c are even. This implies that at least two of a, b, c, d are even. Hence, m is divisible by $2 \cdot 2 \cdot 3 = 12$.

This proves that $m = 12$. ■

Exercise 99 (RMO 2024a P4,). Let $n > 1$ be a positive integer. Call a rearrangement $a_1, a_2, \ldots, a_n$ of $1, 2, \ldots, n$ *nice* if for every $k = 2, 3, \ldots, n$, we have that $a_1^2 + a_2^2 + \cdots + a_k^2$ is not divisible by k. Determine which positive integers $n > 1$ have a nice arrangement.

Solution 94. If n is an odd positive integer and is not divisible by 3, then the sum of the squares of $1, 2, \ldots, n$, that is, the integer

$$\frac{n(n+1)(2n+1)}{6}$$

is a multiple of n. This shows that if $n > 1$ is an integer relatively prime to 6, then $1, 2, \ldots, n$ do not admit any nice rearrangement.

Claim — For any positive integer k, the following rearrangement of $1, 2, \ldots, 6k$ (to be read accross the rows)

$$\begin{array}{cccccc} 2 & 1 & 3 & 4 & 6 & 5 \\ 8 & 7 & 9 & 10 & 12 & 11 \\ 14 & 13 & 15 & 16 & 18 & 17 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 6k-4 & 6k-5 & 6k-3 & 6k-2 & 6k & 6k-1 \end{array}$$

is nice.

The content posted here and at this blog by Evan Chen are quite useful.

Proof of the Claim. Note that

$$\begin{aligned}
\frac{2^2+1^2}{2} &= \frac{(2+1)(2\cdot 2+1)}{6}\\
\frac{2^2+1^2+3^2}{3} &= \frac{(3+1)(2\cdot 3+1)}{6}\\
\frac{2^2+1^2+3^2+4^2}{4} &= \frac{(4+1)(2\cdot 4+1)}{6}\\
\frac{2^2+1^2+3^2+4^2+6^2}{5} &= \frac{(5+1)(2\cdot 5+1)}{6}+\frac{6^2-5^2}{5}\\
\frac{2^2+1^2+3^2+4^2+6^2+5^2}{6} &= \frac{(6+1)(2\cdot 6+1)}{6}
\end{aligned}$$

are not integers. This proves the Claim for $k=1$. Let k be a positive integer such that the Claim holds. Consider the rearrangement (to be read accross the rows)

$$\begin{array}{cccccc}
2 & 1 & 3 & 4 & 6 & 5\\
8 & 7 & 9 & 10 & 12 & 11\\
14 & 13 & 15 & 16 & 18 & 17\\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots\\
6k-4 & 6k-5 & 6k-3 & 6k-2 & 6k & 6k-1\\
6k+2 & 6k+1 & 6k+3 & 6k+4 & 6k+6 & 6k+5
\end{array}$$

of $1,2,\ldots,6k+6$. Note that

$$\frac{\sum_{i=1}^{6k} i^2+(6k+2)^2}{6k+1} = \frac{6k(12k+1)}{6}+\frac{(6k+2)^2}{6k+1}$$

is not an integer since the integers $6k+1, 6k+2$ are relatively prime and $6k+1$ is larger than 1. Observe that

$$\frac{\sum_{i=1}^{6k} i^2+(6k+2)^2+(6k+1)^2}{6k+2} = \frac{(6k+3)(12k+5)}{6}$$

is not an integer since the integers $6k+3, 12k+5$ are odd. Also note that

$$\frac{\sum_{i=1}^{6k} i^2+(6k+2)^2+(6k+1)^2+(6k+3)^2}{6k+3} = \frac{(6k+4)(12k+7)}{6}$$

since 3 does not divide any of $6k+4, 12k+7$. Notice that

$$\frac{\sum_{i=1}^{6k} i^2+(6k+2)^2+(6k+1)^2+(6k+3)^2+(6k+4)^2}{6k+4} = \frac{(6k+5)(12k+9)}{6}$$

is not an integer since $6k+5, 12k+9$ are odd integers. Observe that

$$\frac{\sum_{i=1}^{6k} i^2+(6k+2)^2+(6k+1)^2+(6k+3)^2+(6k+4)^2+(6k+6)^2}{6k+5}$$

$$= \frac{(6k+6)(12k+11)}{6} + \frac{(6k+6)^2-(6k+5)^2}{6k+5}$$

is not an integer since the integers $6k+5, 6k+6$ are relatively prime and $6k+5$ is larger than 1. Also note that

$$\frac{\sum_{i=1}^{6k} i^2 + (6k+2)^2 + (6k+1)^2 + (6k+3)^2 + (6k+4)^2 + (6k+6)^2 + (6k+5)^2}{6k+6}$$
$$= \frac{(6k+7)(12k+13)}{6}$$

is not an integer since the integers $6k+7, 12k+13$ are relatively prime to 6. By induction, the Claim holds for any positive integer k. □

By the above Claim, it follows that if n is a positive integer and n is not relatively prime to 6, then the integers $1, 2, \ldots, n$ admit a nice rearrangement.

This proves that the positive integers $n > 1$, admitting a nice rearrangement, are precisely those which are not relatively prime to 6. ■

Exercise 100 (RMO 2024b P1,). Let $n > 1$ be a positive integer. Call a rearrangement $a_1, a_2, \ldots, a_n$ of $1, 2, \ldots, n$ *nice* if for every $k = 2, 3, \ldots, n$, we have that $a_1 + a_2 + \cdots + a_k$ is not divisible by k.

1. If $n > 1$ is odd, prove that there is no nice arrangement of $1, 2, \ldots, n$.
2. If n is even, find a nice arrangement of $1, 2, \ldots, n$.

Solution 95. Note that if $n > 1$ is an odd positive integer, then for any rearrangement $a_1, a_2, \ldots, a_n$ of $1, 2, \ldots, n$, the integer

$$a_1 + \cdots + a_n = 1 + 2 + \cdots + n = n \times \frac{n+1}{2}$$

is divisible by n. This proves the first part.

For any even positive integer n, a nice rearrangement of $1, 2, \ldots, n$ is provided by the Claim below.

> **Claim —** For any even positive integer n,
>
> $$2, 1, 4, 3, 6, 5, 8, 7, \ldots, n, n-1$$
>
> is a nice rearrangement of $1, 2, \ldots, n$.

Proof of the Claim. Note that the Claim holds for $n = 2$. Let n be an even positive integer satisfying the Claim. Consider the rearrangement

$$2, 1, 4, 3, 6, 5, 8, 7, \ldots, n, n-1, n+2, n+1$$

of $1, 2, \ldots, n+2$. Note that the sum of its first $n+1$ terms is equal to

$$(n+1) \times \frac{n+2}{2} + 1,$$

which is not a multiple of $n+1$. Also note that the sum of the integers $1, 2, \ldots, n+2$ is equal to

$$(n+2) \times \frac{n+3}{2},$$

which is not a multiple of $n+2$. By the induction hypothesis, it follows that the rearrangement

$$2, 1, 4, 3, 6, 5, 8, 7, \ldots, n, n-1, n+2, n+1$$

of $1, 2, \ldots, n+2$ is nice. This proves the Claim. □

■

Exercise 101 (RMO 2024b P2,). For a positive integer n, let $R(n)$ be the sum of the remainders when n is divided by $1, 2, \ldots, n$. For example, $R(4) = 0+0+1+0 = 1$, $R(7) = 0+1+1+3+2+1+0 = 8$. Find all positive integers such that $R(n) = n-1$.

Walkthrough —

(a) Note that

$$\begin{aligned}
R(1) &= 0,\\
R(2) &= 0,\\
R(3) &= 0+1+0\\
&= 1,\\
R(4) &= 0+0+1+0\\
&= 1,\\
R(5) &= 0+1+2+1+0\\
&= 4,\\
R(6) &= 0+0+0+2+1+0\\
&= 3,\\
R(7) &= 0+1+1+3+2+1+0\\
&= 8,\\
R(8) &= 0+0+2+0+3+2+1+0\\
&= 8,\\
R(9) &= 0+1+0+1+4+3+2+1+0\\
&= 12,\\
R(10) &= 0+0+1+2+0+4+3+2+1+0\\
&= 13
\end{aligned}$$

holds.

(b) Does the above help?

Solution 96. For positive integers n and k, denote by $r(n,k)$ the remainder obtained upon dividing n by k. Note that any integer $k \geq 4$,

$$\begin{aligned} R(2k) &\geq r(2k,k-1)+r(2k,k+1)+r(2k,k+2)+\cdots+r(2k,2k-1) \\ &= 2+(k-1)+(k-2)+\cdots+1 \\ &= 2+\frac{k(k-1)}{2} \\ &\geq 2k \end{aligned}$$

holds. Also note that any integer $k \geq 3$,

$$\begin{aligned} R(2k+1) &\geq r(2k+1,k)+r(2k+1,k+1) \\ &\quad +r(2k+1,k+2)+\cdots+r(2k+1,2k-1)+r(2k+1,2k) \\ &= 1+k+(k-1)+\cdots+2+1 \\ &= 1+\frac{k(k+1)}{2} \\ &\geq 2k+1 \end{aligned}$$

holds. Note that

$$\begin{aligned} R(1) &= 0, \\ R(2) &= 0, \\ R(3) &= 0+1+0 \\ &= 1, \\ R(4) &= 0+0+1+0 \\ &= 1, \\ R(5) &= 0+1+2+1+0 \\ &= 4, \\ R(6) &= 0+0+0+2+1+0 \\ &= 3 \end{aligned}$$

holds. This proves that the positive integers n satisfying $R(n) = n-1$ are precisely $1, 5$. ■

Exercise 102 (USAMO 2018 P4, , proposed by Ankan Bhattacharya)**.** Let p be a prime, and let $a_1, a_2, \ldots, a_p$ be integers. Show that there exists an integer k such that the numbers

$$a_1+k, a_2+2k, \ldots, a_p+pk$$

produce at least $p/2$ distinct remainders upon division by p.

The content posted here and at this blog by Evan Chen are quite useful.

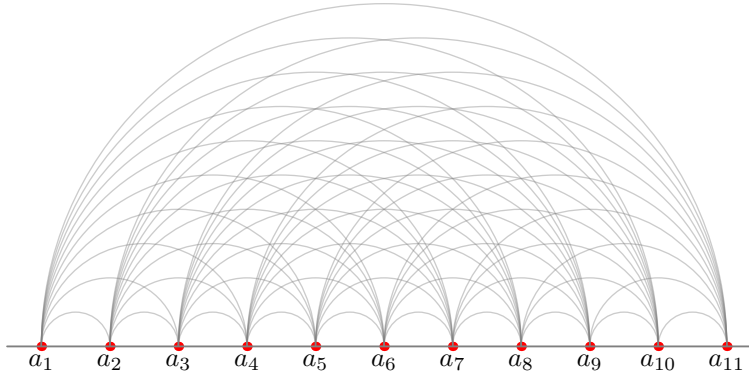


Figure 1: USAMO 2018 P4, Exercise 102

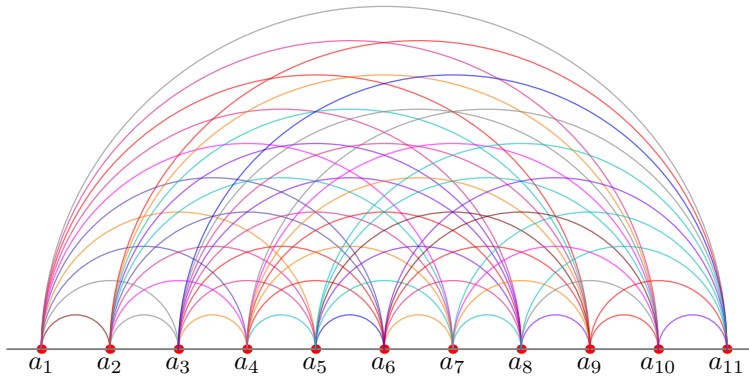


Figure 2: USAMO 2018 P4, Exercise 102

The following uses the fact that every integer not divisible by a prime p is invertible modulo p, which can be proved from first principles!

Walkthrough —

(a) Note that it suffices to show that some integer k among $1, 2, \ldots, p$ has the required property.

(b) Consider the integers $a_1, a_2, \ldots, a_p$, and we join each of them using an arc as in Fig. 1.

(c) For any two distinct integers i, j among $1, 2, \ldots, p$, show that the i-th and the j-th ones among the numbers

$$a_1 + k, a_2 + 2k, \ldots, a_p + pk,$$

that is, the numbers $a_i + ik, a_j + jk$ leave the same remainder upon division by p precisely for one value of k.

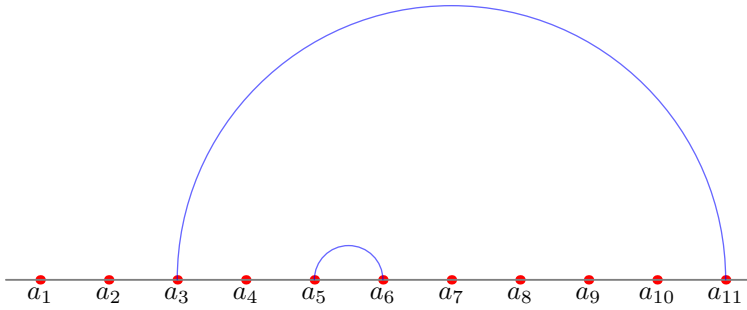


Figure 3: USAMO 2018 P4, Exercise 102

(d) Let us color the arcs (see Fig. 1). We join a_i, a_j by an arc if the numbers $a_i + ik, a_j + jk$ leave the same remainder upon division by p. We write k on that arc. See Fig. 2, where instead of writing the corresponding integers on the arcs, we have colored them. Thinking of the integers $1, 2, \ldots, p$ as p colors, and we have *colored* the arcs instead of *labelling* the arcs[a].

(e) Note that there are precisely $\frac{p(p-1)}{2}$ unordered pairs of the form $\{a_i, a_j\}$, that is, there are precisely $\frac{p(p-1)}{2}$ arcs (see Fig. 1). Moreover, on each of these arcs, one of the integers $1, 2, \ldots, p$ is written. In Fig. 2, these arcs are colored using at most p colors.

(f) Conclude that for some integer k among $1, 2, \ldots, p$, at most $(p-1)/2$ arcs have the integer k written on them, that is, there are at most $(p-1)/2$ unordered pairs of the form $\{a_i, a_j\}$ such that $a_i + ik, a_j + jk$ produce the same remainder upon division by p. In Fig. 2, for some color, there are at most $(p-1)/2$ arcs of that color.

(g) Show that[b] for this integer k, the numbers

$$a_1 + k, a_2 + 2k, \ldots, a_p + pk$$

produce at least $p/2$ distinct remainders upon division by p.

[a] Observe that all the p colors might not have been used, for instance, if any two of $a_1, \ldots, a_p$ differ by a multiple of p.

[b] A graph on n vertices with m edges has at least $n - m$ connected components. This can be proved by induction on m, and observing that introducing a new edge reduces the number of connected components at most by 1.

Exercise 103 (IOQM 2024 P8,). Let n be the smallest integer such that the sum of the digits of n is divisible by 5, as well as the sum of the digits of $n + 1$ is divisible by 5. What are the first two digits of n in the same order?

Summary — Consider the last digit of n. Next consider its second last digit, the third last digit etc.

Walkthrough —

(a) Show that the last digit of n is 9.

(b) Note that n has at least two digits. If n is a two-digit number, then show that it is not equal to any integer other than 19 or 69.

(c) Conclude that n has at least three digits, and its second last digit is 9.

(d) If n is a three-digit number, then show that it is not equal to any integer other than 299 or 799.

(e) Argue similarly to show that n has at least four digits, and its third last digit is 9.

(f) Prove that n has at least five digits, and its fourth last digit is 9.

(g) If n is a five-digit number, then show that it is equal to 49999.

Do we need to check that 49999 has the required property?

Solution 97. ■

Exercise 104 (British Mathematical Olympiad Round 1 2008/9 P4)**.** Find all positive integers n such that both $n + 2008$ divides $n^2 + 2008$ and $n + 2009$ divides $n^2 + 2009$.

Walkthrough —

(a)

Solution 98. Let n be a positive integer such that both $n + 2008$ divides $n^2 + 2008$ and $n + 2009$ divides $n^2 + 2009$. Note that if a is a positive integer such that $n + a$ divides $n^2 + a$, then noting that

$$\begin{aligned} n^2 + a &= (n + a)^2 - 2na - a^2 + a \\ &= (n + a)^2 - 2a(n + a) + a^2 + a, \end{aligned}$$

it follows that $n + a$ divides $a^2 + a = a(a + 1)$. Thus, $n + 2008$ divides

$$2008 \cdot 2009 = 2^3 \cdot 7^2 \cdot 41 \cdot 251,$$

and $n + 2009$ divides

$$2009 \cdot 2010 = 2 \cdot 3 \cdot 5 \cdot 7^2 \cdot 41 \cdot 67.$$

Let us consider the case that 251 does not divide $n + 2008$. Since $n + 2008$ is greater than $2^3 \cdot 7^2$, it follows that 41 divides $n + 2008$. If 7 does not

divide $n+2008$, then $n+2008$ divides $2^3 \cdot 41$, which is not possible since $n+2008 \geq 2009 = 7^2 \cdot 41 > 2^3 \cdot 41$. Thus, 7 divides $n+2008$. Since the integers $n+2008, n+2009$ have no factor in common other than 1, it follows that $n+2009$ divides $2 \cdot 3 \cdot 5 \cdot 67 = 2010$, which implies that $n = 1$.

It remains to consider the case that 251 divides $n+2008$, which we assume henceforth. Note that $n+2009$ is not divisible by 4, and hence, $n \not\equiv 3 \pmod 4$. Let us consider the case that n is odd. Then, $n+2008$ is odd, and hence, $n+2008$ divides $7^2 \cdot 41 \cdot 251$. Using $n+2008 \equiv 1 \pmod 4$, $251 \equiv 3 \pmod 4$ and $41 \equiv 1 \pmod 4$, it follows that 7 divides $n+2008$. Using $n+2008 \geq 2009$ and $n+2008 \equiv 1 \pmod 4$, we obtain that $n+2008$ is equal to none of

$$7 \cdot 251, 7^2 \cdot 251.$$

Hence, $n+2008$ is divisible by 7 and 41. Since the integers $n+2008, n+2009$ have no factor in common other than 1, it follows that $n+2009$ divides $2 \cdot 3 \cdot 5 \cdot 67 = 2010$, which implies that $n = 1$. It remains to consider the case that n is even, which we assume henceforth. Since $n+2008$ is greater than $2^3 \cdot 251$, it follows that $n+2008$ is divisible by at least one of $7, 41$. If each of $7, 41$ divides $n+2008$, then since the integers $n+2008, n+2009$ have no factor in common other than 1, it follows that $n+2009$ divides $3 \cdot 5 \cdot 67 = 1005$, which is impossible since $n+2009 \geq 2010$. Thus, exactly one of $7, 41$ divides $n+2008$.

Let us consider the case that 7 divides $n+2008$. Then $n+2009$ divides $3 \cdot 5 \cdot 41 \cdot 67$. Since $n+2009 \geq 2010$, it follows that 41 and 67 divide $n+2009$. Note that

$$41 \equiv -1 \pmod 7, \quad 67 \equiv -3 \pmod 7$$

holds, implying that $41 \cdot 67 \equiv 3 \pmod 7$. Since $n+2009 \equiv 1 \pmod 7$, we obtain $n+2009 = 5 \cdot 41 \cdot 67$, which yields

$$n+2008 \equiv 4-1 \equiv 0 \pmod 3.$$

Thus, $n+2008$ is divisible by 3, which is a contradiction. It remains to consider the case that 41 divides $n+2008$, which we assume henceforth. Note that $n+2009$ divides $3 \cdot 5 \cdot 7^2 \cdot 67$. Since $n+2009 \geq 2010$, it follows that 7 divides $n+2009$. This gives $n+2008 \equiv -1 \pmod 7$. Note that $41 \cdot 251 \equiv 41^1 \equiv 1 \pmod 7$ holds. Since $n+2008$ is a multiple of $2 \cdot 41 \cdot 251$, and divides $2^3 \cdot 41 \cdot 251$, it follows that one of $2, 2^2, 2^3$ is congruent to -1 modulo 7, which is a contradiction.

Therefore, there is only one positive integer n such that both $n+2008$ divides n^2+2008 and $n+2009$ divides n^2+2009, and that is $n = 1$. ■

Exercise 105 (British Mathematical Olympiad Round 1 2014/15 P2)**.** Positive integers a, b and prime p satisfy the equation

$$p^2 + a^2 = b^2.$$

Prove that if p is a prime greater than 3, then a is a multiple of 12 and that $2(p+a+1)$ is a perfect square.

Walkthrough —

(a)

Solution 99. Assume that p is a prime greater than 3. If a is not divisible by 3, then

$$p^2 + a^2 \equiv 2 \pmod 3$$

holds, which implies that b^2 is congruent to 2 modulo 3. Since b^2 is congruent to either 0 or 1 modulo 3, it follows that a is divisible by 3. If a is odd, then a^2 is congruent to 1 modulo 8, which implies that b^2 is congruent to 2 modulo 4, which is impossible. This implies that a is even. If a is congruent to one of $2, -2$ modulo 8, then a^2 is congruent to 4 modulo 8, which implies that b^2 is congruent to 5 modulo 8. Since b^2 is congruent to either $0, 1, 4$ modulo 8, it follows that a is congruent to one of $0, 4$ modulo 8. Using the fact that a is divisible by 3 and 4, we conclude that a is a multiple of 12.

Note that

$$(b - a)(b + a) = p^2$$

holds. Since p is a prime and $1 < b - a < b + a$ holds, we have $b - a = p^2$ and $b + a = 1$, which yields

$$2a = p^2 - 1.$$

We obtain

$$2(p + a + 1) = 2p + p^2 - 1 + 2 = (p + 1)^2.$$

This shows that $2(p + a + 1)$ is a perfect square. ■

Exercise 106 (RMO 2025a P4,). Prove that there do not exist positive rational number x and y such that

$$x + y + \frac{1}{x} + \frac{1}{y} = 2025.$$

Walkthrough —

(a) Consider the problem of showing that the given equation has no solutions in positive integers.

(b) Determine and compare the denominators of both sides of

$$x + \frac{1}{x} = 2025 - y - \frac{1}{y}.$$

(c) Does the above provide any insight into the original problem?

Solution 100. On the contrary, suppose there exist positive rational numbers x and y such that

$$x + y + \frac{1}{x} + \frac{1}{y} = 2025.$$

Let a, b, c, d be positive integers such that $x = \frac{a}{b}$ and $y = \frac{c}{d}$, where $\gcd(a, b) = 1$ and $\gcd(c, d) = 1$. Note that

$$x + \frac{1}{x} = \frac{a^2 + b^2}{ab}, \quad \text{and} \quad y + \frac{1}{y} = \frac{c^2 + d^2}{cd}.$$

Observe that

$$\frac{a^2 + b^2}{ab} = 2025 - \frac{c^2 + d^2}{cd} = \frac{2025cd - (c^2 + d^2)}{cd},$$

with $\gcd(a^2 + b^2, ab) = 1$ and

$$\gcd(2025cd - (c^2 + d^2), cd) = \gcd(c^2 + d^2, cd) = 1.$$

This implies that

$$ab = cd.$$

It follows that

$$a^2 + b^2 + c^2 + d^2 = 2025cd.$$

Using the integers a, b have no common prime factors, it follows that at most one of a and b is divisible by 3. Similarly, at most one of c and d is divisible by 3. Therefore, among the four integers a, b, c, d, at most two of them are divisible by 3. If none of them is divisible by 3, then

$$a^2 + b^2 + c^2 + d^2 \equiv 4 \pmod 3 \equiv 1 \pmod 3,$$

which is impossible since 3 divides 2025. This shows that at least one of a, b, c, d is divisible by 3. Using $ab = cd$, it follows that at least two of a, b, c, d are divisible by 3. Using the above, we obtain that exactly two of a, b, c, d are divisible by 3. This implies that

$$a^2 + b^2 + c^2 + d^2 \equiv 2 \pmod 3,$$

which is again impossible since 3 divides 2025.

This proves that there do not exist positive rational numbers x and y such that

$$x + y + \frac{1}{x} + \frac{1}{y} = 2025.$$

■

§1.6 More on congruences

The content posted here and at this blog by Evan Chen are quite useful.

Theorem 1 (Wilson)

For any prime p, the congruence

$$(p-1)! \equiv -1 \pmod{p}$$

holds.

Example 107. Let $n \geq 5$ be an integer. Show that n is composite if and only if n divides $(n-1)!$.

Solution 101. If n is a prime, then by Wilson's theorem, it follows that n does not divide $(n-1)!$. This proves the "if part".

To prove the "only if part", note that if n is composite, then the highest power of any of its prime divisors is at most $n-1$, and hence divides $(n-1)!$, and since these powers are pairwise coprime, it follows that n divides $(n-1)!$. ■

Theorem 2 (Fermat's little theorem)

If p is prime, and a is an integer not divisible by p, then

$$a^{p-1} \equiv 1 \pmod{p}$$

holds.

Example 108. Show that for every prime number p, there exists an integer $n > 1$ such that

$$2^n + 3^n + 6^n - 1$$

is divisible by p.

Summary — Observe that

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1,$$

and use Fermat's little theorem.

Walkthrough — quad

(a) If $p = 2$ or $p = 3$, then $n = 2$ works.

(b) If $p > 3$ holds, then does there exist a positive integer such that $2^n, 3^n, 6^n$ can **thought of** as $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ modulo p respectively?

Solution 102. If $p = 2$ or $p = 3$, then $n = 2$ works. Now, let us assume that $p > 3$. Since p does not divide any of $2, 3, 6$, by Fermat's little theorem, it follows that

$$2^{p-1} \equiv 1 \pmod{p}, \quad 3^{p-1} \equiv 1 \pmod{p}, \quad 6^{p-1} \equiv 1 \pmod{p}.$$

This shows that $n = p - 2$ works. Indeed, the above congruences imply that

$$6(2^{p-2} + 3^{p-2} + 6^{p-2} - 1) \equiv 3 + 2 + 1 - 6 \pmod{p}$$

holds. Since p are 6 are relatively prime, it follows that

$$2^{p-2} + 3^{p-2} + 6^{p-2} - 1 \equiv 0 \pmod{p}.$$

■

Exercise 109 (Estonia Math Olympiad 2025 Open Contest O14,). Is there a positive integer n such that 88 divides $2^n + n^3$?

> **Walkthrough** — Consider congruences modulo 11.

Exercise 110 (USAJMO 2013 P1, proposed by Titu Andreescu). Are there integers a, b such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes of integers?

> **Summary** — Consider the integer ab modulo 3, and read $a^5b + 3, ab^5 + 3$ modulo 9.

> **Walkthrough** —
>
> **(a)** If 3 divides ab, then consider one of the integers $a^5b + 3, ab^5 + 3$ modulo 9.
>
> **(b)** If 3 does not divide ab, then assuming the integers $a^5b + 3, ab^5 + 3$ to be perfect cubes, determine the integers a^5b, ab^5 modulo 9.
>
> **(c)** Consider the integer $(ab)^6$ modulo 9.

Solution 103. Let us assume that there are integers a, b satisfying the given condition. First, let us consider that 3 divides ab. Without loss of generality[3], let us assume that 3 divides a. Then

$$a^5b + 3 \equiv 3 \pmod{9}.$$

Since $a^5b + 3$ is a perfect cube and is divisible by 3, it follows that

$$a^5b + 3 \equiv 0 \pmod{9},$$

[3]Show that there is indeed no loss of generality in assuming what follows.

The content posted here and at this blog by Evan Chen are quite useful.

implying $3 \equiv 0 \pmod 9$, which is impossible.

Next, let us consider the case that 3 does not divide ab. Since $a^5b + 3$ is a cube and is not divisible by 3, we obtain $a^5b + 3 \equiv \pm 1 \pmod 9$, which implies that a^5b is congruent to one of $5, 7$ modulo 9. Using a similar argument, it follows that ab^5 is also congruent to one of $5, 7$ modulo 9. Hence, $(ab)^6$ is congruent to one of $5 \cdot 5, 5 \cdot 7, 7 \cdot 7$ modulo 9. So, $(ab)^6$ is not congruent to 1 modulo 9. Since 3 does not divide ab, it follows that $(ab)^6$ is congruent to 1 modulo 9.

It follows that there are no integers a, b satisfying the given conditions. ■

Exercise 111 (Austrian Mathematical Olympiad, Regional Competition 2024 P4, , proposed by Walther Janous,). Let n be a positive integer. Prove that $a(n) = n^5 + 5^n$ is divisible by 11 if and only if $b(n) = n^5 \cdot 5^n + 1$ is divisible by 11.

> **Walkthrough** — Does $n^5 \cdot a(n)$ agree with $b(n)$ modulo 11 under some conditions on n?

Exercise 112 (IMOSL 1984 P2). Prove that if m, n are positive integers, then $4mn - m - n$ cannot be a perfect square.

> **Walkthrough** —
>
> **(a)**

Solution 104. Let m, n be positive integers such that $4mn - m - n = k^2$ holds for some positive integer k^2. This yields

$$(4m - 1)(4n - 1) = 4k^2 + 1.$$

Since $4m - 1$ is a positive integer and it is congruent to 3 modulo 4, it admits a prime divisor p satisfying $p \equiv 3 \bmod 4$. Note that p divides $4k^2 + 1$, and p does not divide $2k$. Applying Fermat's little theorem, it follows that

$$(-1)^{\frac{p-1}{2}} \equiv ((2k)^2)^{\frac{p-1}{2}} \equiv 1 \bmod p,$$

which is impossible since $p \equiv 3 \bmod 4$. ■

Exercise 113 (RMO 1990 P4,). Find the remainder when 2^{1990} is divided by 1990.

Solution 105. Since 199 is a prime, by Fermat's little theorem, we obtain

$$2^{198} \equiv 1 \bmod 199,$$

which yields

$$2^{1990} \equiv 2^{1980} \cdot 2^{10} \bmod 199 \equiv 29 \bmod 199.$$

Using $2^4 \equiv 1 \bmod 5$, we get

$$2^{1990} \equiv 4 \bmod 5 \equiv 29 \bmod 5.$$

Since $5, 199$ are coprime, this shows that

$$2^{1990} \equiv 29 \bmod 199 \cdot 5.$$

Observe that

$$2^{1990} - (29 + 199 \cdot 5)$$

is divisible by $199 \cdot 5$, and also by 2. Since $2, 199 \cdot 5$ are coprime, it follows that

$$2^{1990} \equiv 29 + 199 \cdot 5 \bmod 2 \cdot 199 \cdot 5,$$

which implies that

$$2^{1990} \equiv 1024 \bmod 1990.$$

So, one obtains 1024 as the remainder upon dividing 2^{1990} by 1990. ■

Exercise 114 (RMO 1990 P6,). N is a 50 digit number (in the decimal scale). All digits except the 26th digit (from the left) are 1. If N is divisible by 13, find the 26th digit.

Solution 106. Let a denote the 26-th digit of N. The given condition yields

$$N = 1 + 10 + 10^2 + \cdots + 10^{49} + (a-1)10^{50-26}.$$

Assume that N is divisible by 13. This gives

$$1 + 10 + 10^2 + \cdots + 10^{49} + (a-1)10^{24} \equiv 0 \bmod 13.$$

Applying Fermat's little theorem, we obtain $10^{12} \equiv 1 \bmod 13$, and this shows that

$$10^{24} \equiv 1 \bmod 13, \quad 10^{50} - 1 \equiv 10^2 - 1 \bmod 13.$$

Note that

$$9(1 + 10 + 10^2 + \cdots + 10^{49}) + 9(a-1)10^{24} \equiv 0 \bmod 13$$

holds, which shows that

$$10^{50} - 1 + 9(a-1)10^{24} \equiv 0 \bmod 13,$$

and this implies that

$$10^2 - 1 + 9(a-1) \equiv 0 \bmod 13.$$

Since the integers $9, 13$ are relatively prime, we obtain

$$11 + a - 1 \equiv 0 \bmod 13.$$

It follows that $a = 3$. ■

Theorem 3 (Euler)

Let n be a positive integer, and $\varphi(n)$ denote the number of integers lying between 1 and n which are relatively prime to n. If a is an integer relatively prime to n, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

holds.

Example 115. Let a, b be relatively prime positive integers. Show that there exist positive integers m, n such that

$$a^m + b^n \equiv 1 \pmod{ab}.$$

Walkthrough — Read it modulo a and modulo b, and use Euler's theorem.

Solution 107. Since a, b are relatively prime, it suffices to show that there exist positive integers m, n such that

$$a^m \equiv 1 \pmod{b}, \quad b^n \equiv 1 \pmod{a}.$$

By Euler's theorem, it follows that

$$a^{\varphi(b)} \equiv 1 \pmod{b}, \quad b^{\varphi(a)} \equiv 1 \pmod{a},$$

and hence, the required result follows by taking $m = \varphi(b), n = \varphi(a)$. ■

Exercise 116 (Kosovo National Olympiad 2023 Grade 12 P4,). Let p be an odd prime number. Prove that for every such p, regardless of how we divide the set

$$\mathbb{A} = \{1^{p^2-p}, 2^{p^2-p}, \ldots, (p-1)^{p^2-p}, p^2 - p + 1\},$$

into two disjoint sets, the arithmetic means of the elements of the two subsets cannot be equal.

Walkthrough —

(a)

Solution 108. Let S and T be the two disjoint subsets of $\mathbb{A}$ such that $S \cup T = \mathbb{A}$. Write $S = \{a_1, a_2, \ldots, a_k\}$ and $T = \{b_1, b_2, \ldots, b_{p-k}\}$, where $1 \leq k \leq p-1$.

We want to show that the arithmetic means of the elements of S and T are not equal. On the contrary, suppose that they are equal, that is,

$$\frac{a_1 + a_2 + \cdots + a_k}{k} = \frac{b_1 + b_2 + \cdots + b_{p-k}}{p-k},$$

which yields

$$(p-k)(a_1 + a_2 + \cdots + a_k) = k(b_1 + b_2 + \cdots + b_{p-k}).$$

This gives

$$\begin{aligned} &p(a_1 + a_2 + \cdots + a_k) \\ &= k(a_1 + a_2 + \cdots + a_k + b_1 + b_2 + \cdots + b_{p-k}) \\ &= k(1^{p^2-p} + 2^{p^2-p} + \cdots + (p-1)^{p^2-p} + (p^2 - p + 1)). \end{aligned}$$

Since p is a prime, using Euler's theorem, we have $i^{p^2-p} \equiv 1 \pmod{p^2}$ for all $1 \le i \le p-1$, which gives

$$1^{p^2-p} + 2^{p^2-p} + \cdots + (p-1)^{p^2-p} + (p^2 - p + 1) \equiv 0 \pmod{p^2}.$$

This shows that p divides $a_1 + a_2 + \cdots + a_k$. Note that the elements of S are congruent to 1 modulo p. This implies $a_1 + a_2 + \cdots + a_k \equiv k \pmod{p}$, which is a contradiction since $1 \le k \le p-1$. Consequently, the arithmetic means of the elements of S and T are not equal. ■

Exercise 117 (RMO 1993 P5,). Show that $19^{93} - 13^{99}$ is a positive integer divisible by 162.

Solution 109. Note that the congruences

$$\begin{aligned} 19^{93} &\equiv (1+18)^{93} \bmod 9^2 \\ &\equiv 1 + 93 \cdot 18 \bmod 9^2 \\ &\equiv 1 + 12 \cdot 18 \bmod 9^2 \\ &\equiv 1 + 8 \cdot 27 \bmod 9^2 \\ &\equiv 1 - 27 \bmod 9^2 \\ &\equiv -26 \bmod 9^2 \\ &\equiv 55 \bmod 9^2, \\ 13^{99} &\equiv (4+9)^{99} \bmod 9^2 \\ &\equiv 4^{99} + 99 \cdot 4^{98} \cdot 9 \bmod 9^2 \\ &\equiv 4^{99} \bmod 9^2 \end{aligned}$$

hold. Also note that

$$\begin{aligned} 4^{81} &\equiv 1 + 81 \cdot 3 + \binom{81}{2} 3^2 + \binom{81}{3} 3^3 \bmod 9^2 \\ &\equiv 1 \bmod 9^2 \end{aligned}$$

holds, which yields

$$\begin{aligned} 4^{99} &\equiv 4^{81} \cdot 4^{18} \bmod 81 \\ &\equiv 4^{18} \bmod 81 \\ &\equiv (1+3)^{18} \bmod 81 \\ &\equiv 1 + 18 \cdot 3 + \frac{18 \cdot 17}{2} 3^2 + \frac{18 \cdot 17 \cdot 16}{6} 3^3 \bmod 81 \\ &\equiv 1 + 18 \cdot 3 \bmod 81 \\ &\equiv 55 \bmod 81. \end{aligned}$$

It follows that $19^{93} - 4^{99}$ is divisible by 81, and hence, $19^{93} - 13^{99}$ is divisible by 81. Since $19^{93} - 13^{99}$ is an even number, and $2, 81$ are coprime, we conclude that $19^{93} - 13^{99}$ is divisible by 162. ■

Exercise 118 (RMO 2015f P3)**.** Let

$$N = 2^5 + 2^{5^2} + 2^{5^3} + \cdots + 2^{5^{2015}},$$

written in the usual decimal form, find the last two digits of the number N.

Solution 110. Applying Euler's theorem, we obtain [4]

$$2^{20} \equiv 2^{\varphi(25)} \equiv 1 \bmod 25.$$

Note that for any integer $n \geq 1$,

$$5^n - 5 = 5(5^{n-1} - 1) = 5((1+4)^{n-1} - 1)$$

holds, and hence we obtain $5^n \equiv 5 \bmod 20$. This yields

$$N = 2^5 \left(\sum_{i=1}^{2015} 2^{5^i - 5} \right) = 2^5 \left(\sum_{i=1}^{2015} (2^{20})^{\frac{5^i - 5}{20}} \right),$$

and consequently, we obtain

$$N \equiv 2^5 \cdot 2015 \equiv 7 \cdot 15 \equiv 5 \bmod 25.$$

Also note that $N \equiv 0 \bmod 4$ holds. It follows that $N - 80$ is divisible by the relatively prime integers $4, 25$, which shows that 100 divides $N - 80$. Hence, the last two digits of N are $8, 0$. ■

[4] Alternatively, one may observe that $2^{10} = 1024 \equiv -1 \bmod 25$, which yields $2^{20} \equiv 2^{\varphi(25)} \equiv 1 \bmod 25$.

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